

Skill Representation for Scalable Agent Skill Retrieval under Semantic Confusability

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Abstract

Large language model agents increasingly rely on reusable skills that package instructions, work-flows, resources, dependencies, and constraints. The core difficulty is selecting the correct skill when several share a topic but differ in inputs, outputs, procedures, resources, or authority boundaries.

This thesis asks which operational information distinguishes semantically similar but procedurally distinct skills, and how representation and retrieval choices preserve or use it. Candidate fields are motivated by the literature and by a full-document binary review of 29,292 public skills, which finds every candidate type common but not universal. RQ1 combines 350 controlled three-sibling clusters with source-grounded information removal from 1,099 public originals. RQ2a holds queries, candidates, and operational propositions fixed while comparing matched fielded, flat, prose, order-controlled, and diluted representations. RQ2b compares native metadata, complete bodies, unlabelled evidence, and field-labelled evidence across a 2,433-skill library and 381 strict-gold prompts.

All seven RQ1 fields resolve deliberately tied near-neighbours in the controlled setting, although their robustness varies by prompt and retriever. Examples/tests are a negative control: generic examples support execution without defining distinct capabilities. In public originals, removing use-condition or success/verification evidence lowers strict Hit@1 under both BM25 and Qwen; other individual fields are retriever-conditional. RQ2a shows that restoring candidate-specific operational facts improves every selector over a shared-only control, whereas explicit headings are not universally beneficial: versus matched flat text, they improve BM25 top-1 by 2.1 points, reduce Qwen single-vector top-1 by 6.7 points, and change SkillRouter cross-encoder top-1 by -0.4 points.

On RQ2b’s public-gold stratum, complete bodies exceed field-labelled evidence in first-stage top-1 for BM25 (0.644 versus 0.504), Qwen (0.686 versus 0.538), and SkillRouter (0.806 versus 0.746), while field-labelled evidence reduces selector-visible corpus tokens by 72–76% and document/index time by 64–79%. The thesis therefore supports an accuracy–compression trade-off, not a universal advantage for field labels or a claim that field presence alone improves retrieval.

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Introduction

1.1 Background

Large language model (LLM) agents are increasingly designed as systems that combine reasoning, action, memory, and external capability use rather than as standalone text generators. Early work on retrieval-augmented generation showed that language models can improve performance by accessing external information at inference time rather than relying solely on knowledge stored in model parameters [4, 6]. Later work on tool use extended this idea from document retrieval to action execution by enabling models to call external functions and APIs [14, 15, 17].

More recently, agent systems have begun to treat reusable capabilities as *skills*. A skill is not simply an atomic tool call. It may include instructions, workflow steps, scripts, examples, dependencies, input requirements, output expectations, and constraints. In practical systems such as filesystem-based agent skill libraries, the full skill artifact may be stored as a directory containing a `SKILL.md` file plus optional resources. The agent may initially see only a compressed representation, such as a name and description, before deciding whether to load the full artifact.

This shift changes the retrieval problem. If the object being selected is a high-level procedure rather than an atomic function, retrieval quality depends not only on topical relevance, but also on procedural suitability. Two skills may look semantically similar because they refer to the same domain or task family, while still differing in preconditions, workflow steps, required resources, output format, or side effects.

1.2 Problem Statement

The problem addressed in this thesis is semantic confusion in large agent skill libraries. As skill libraries grow, a user request may match many plausible skills at the level of names and descriptions. The correct skill may exist, but the retrieval system may surface a nearby skill that sounds relevant while being procedurally wrong.

This failure mode is different from ordinary retrieval failure. The wrong result is often not unrelated. It is close enough to be tempting. For example, a request to “extract information from this PDF” may plausibly match document review, document summarisation, citation grounding, field extraction, or layout-preserving conversion. These skills may all mention documents, extraction, or PDFs, but they require different procedures and produce different outputs. A flat description may not preserve the procedural distinctions required for correct selection.

The scalability issue makes this worse. In a small library, the main agent may be able to inspect all skill names and short descriptions in context. In a larger library, even this metadata can become too large or noisy. The system then needs a candidate-subsetting stage before the main agent loads any full skill document. This makes the skill representation critical: the correct skill must be surfaced before the full artifact can be inspected.

1.3 Research Questions

This thesis uses the following working research questions:

RQ1: Which operational information types help distinguish semantically similar but procedurally distinct agent skills?

RQ2: How do skill representation and retrieval-pipeline choices affect the preservation and use of the routing-relevant operational information identified in RQ1 under semantic confusability, and what accuracy, candidate-recall, and retrieval-cost trade-offs result?

These questions deliberately narrow the thesis from a broad comparison of “flat” and “structured” skill representation and retrieval to a diagnostic study of routing-relevant information. RQ1 asks which kinds of operational information actually make near-neighbour skills distinguishable. RQ2 separates a matched-content mechanism study from a full-library validation stage. RQ2a fixes the query, three

sibling candidates, reviewed operational propositions, and selector while varying how the same facts are organised, flattened, expressed as prose, diluted, or explicitly scored field by field. RQ2b then evaluates four source-grounded representations—I1 native name–description metadata, I2 complete skill bodies, I3-flat extracted evidence, and field-labelled I3C—over a 2,433-skill library and 381 strict-gold prompts. It separates three first-stage retrievers from two fixed-Top-20 rerankers, so candidate inclusion, final ranking, and measured selector cost remain distinguishable.

This framing also limits the novelty claim. The thesis does not propose a new universal schema for agent skills. Recent structured skill representations such as Scheduling-Structural-Logical (SSL) already decompose skill artifacts into invocation-level signals, execution structure, and action/resource evidence. The contribution here is diagnostic and empirical: given semantically similar candidate skills, it tests which operational fields make the correct skill distinguishable and which skill retrieval methods can use those fields reliably.

1.4 Scope

The thesis focuses on retrieval and candidate selection before full skill execution. It does not attempt to build a complete general-purpose agent operating system. The main agent, the retriever, and the reranker are treated as separable components. The retriever may be a lexical method, an embedding model, a deterministic schema scorer, a neural reranker, or a hybrid pipeline. What matters is whether the retrieval stage can produce a candidate set that contains and ranks the procedurally correct skill.

The evaluation focuses on skill libraries where some skills are intentionally confusable. This is important because broad relevance retrieval is too easy for the research question. A benchmark that only asks whether a document task retrieves a document skill would not test the central failure mode. The difficult case is selecting between skills that share the same broad domain but differ in operational requirements.

1.5 Contributions

This thesis makes five contributions:

- (1) A field-isolated benchmark design for skill retrieval under semantic confusability and procedural distinctness, with controlled near-neighbour clusters that vary one operational signal at a time.

- (2) Empirical evidence about which operational information types act as routing signals, distinguishing strong positive routing signals, conditional or tie-breaker signals, and information that mainly supports execution after selection.
- (3) A matched-content RQ2a comparison that separates operational-fact content, explicit field organisation, prose form, information dilution, and selector behaviour under lexical, single-vector semantic, cross-encoder, and semantic field-aware selection.
- (4) A source-grounded RQ2b full-library validation over 2,433 skills and 381 strict-gold prompts, comparing I1, I2, I3-flat, and I3C under lexical, Qwen, and SkillRouter first-stage retrieval plus two fixed-Top-20 rerankers; it reports an accuracy–compression trade-off rather than a universal I3C gain.
- (5) A failure-mode and efficiency analysis of when routing-relevant information is absent, diluted, implicit, misleading, or present but not exploited, while treating tree/graph architectures and downstream execution as future work.

The contribution is therefore positioned as a controlled analysis of discriminative skill information, not as the first proposal of structured skill representation.

1.6 Thesis Structure

Chapter 2 reviews the literature on retrieval-augmented language models, tool use, agent skills, skill-library scaling, and structure-aware retrieval. Chapter 3 defines the vocabulary and layer distinctions used throughout the thesis. Chapter 4 describes the benchmark design. Chapter 5 explains the methodology and evaluation metrics. Chapter 6 reports the results. Chapter 7 discusses implications and failure modes. Chapter 8 concludes the thesis and outlines future work.

Literature Review

2.1 Skills as a Distinct Retrieval Target

Skills should be treated as a distinct retrieval target, rather than as ordinary documents or larger tools. Documents are retrieved because they contain evidence that may help a model answer a question. Atomic tools are selected because they expose a particular callable function and argument interface. Skills sit between and beyond these cases. Like documents, they may contain natural-language instructions, examples, and reference material. Like tools, they may invoke scripts, APIs, resources, or other capabilities. However, their primary unit is neither a passage nor a single function call. A skill is a reusable procedural capability: it describes when a capability should be invoked, what input or precondition it expects, what workflow it follows, what dependencies or resources it uses, and what output or success condition it is meant to produce [1, 5, 21].

This procedural character is central in recent skill-focused literature. The procedural-memory survey frames skills as reusable memories of how to perform tasks, organised around acquisition, representation, invocation, and refinement [21]. The SoK on agentic skills similarly argues that skills are more than tools, plans, or memory items because they bundle executable procedural knowledge into reusable modules [5]. These accounts suggest that skill selection should be understood as procedural suitability rather than topical relevance alone. A retriever that selects a document only needs to find material relevant enough to support the next reasoning step. A retriever that selects a tool must match an intended function and its arguments. A retriever that selects a skill must identify a procedure whose applicability, workflow, dependencies, outputs, and boundaries fit the user’s intent. A selected skill can therefore be semantically relevant and still be procedurally wrong.

Early skill systems show why this distinction matters. SkillAct represents skills as textual abstractions containing a name, natural-language instructions, and example trajectories, and shows that such abstractions can improve agent performance in embodied tasks [13]. Its design demonstrates that reusable procedural units can help an agent avoid repeatedly solving the same subproblem from scratch. However, SkillAct largely exposes the relevant skills directly to the agent and does not directly evaluate selection among many overlapping skills as the library grows. Voyager offers a stronger example of an expanding skill library: it stores successful behaviours as executable code and retrieves them from a library indexed by natural-language descriptions [20]. This demonstrates the usefulness of persistent external skills, but it also illustrates the limitation of description-level retrieval. A description can identify a rough capability area, yet fail to preserve the operational distinctions needed when several skills use similar domain language but differ in input state, procedure, dependency, or output.

Filesystem-based skill libraries make the same issue concrete. A complete skill artifact may be a directory containing a `SKILL.md` file, scripts, templates, reference files, examples, or dependency instructions [1]. The selector, however, may initially see only a compressed representation such as a name and short description. This creates a layered retrieval problem: the stored artifact may contain rich procedural information, but skill retrieval may not see or use it. As a hypothetical illustration, consider three skills that could all be advertised as “extract financial tables from a PDF”. One expects a born-digital annual report and uses the text layer plus table geometry to produce a CSV with page and cell coordinates. Another expects scanned statements and performs OCR cleanup before returning normalised JSON. A third takes an already extracted table and verifies each number against page-grounded evidence, returning an audit report rather than a fresh extraction. At the short-description level, all three concern financial tables, PDFs, extraction, and evidence. They are nevertheless not interchangeable, because they differ in input state, workflow, output artifact, dependencies, success criteria, and boundaries.

This distinction motivates the thesis terminology. A *skill artifact* is the stored capability itself, such as a `SKILL.md` file, scripts, resources, examples, or generated program. A *skill representation* is the selector-visible form of that skill, such as a short description, full text body, structured card, hierarchy node, or graph neighbourhood. *Skill retrieval* is the process that uses those representations to produce candidates, rerank them, traverse an index, or load fuller skill instructions. Keeping these terms separate matters because a result may come from better artifact content, a richer representation, a stronger first-stage retriever, a reranker, or a more favourable evaluation setup. This thesis therefore treats skill

retrieval as the problem of selecting a reusable procedural artifact through a representation that may or may not preserve the information required for correct invocation.

2.2 Scale and Semantic Confusability

Progressive disclosure is a useful practical response to context pressure. Anthropic’s skill documentation describes filesystem skill artifacts that may include a `SKILL.md` file, scripts, resources, references, and dependency instructions, while the model initially sees compact skill metadata and loads the full artifact only when the skill appears relevant [1]. This avoids placing every full skill document directly in context. However, it does not remove the large-library routing problem. In a straightforward discovery setting, the agent still needs access to available skill names and descriptions so that it can decide which full artifact to load. As the library grows, even this compact discovery layer can consume context, increase prompt cost, slow selection, and create more opportunities for plausible but incorrect matches.

Benchmark evidence shows that scale is a real pressure in skill libraries. SkillsBench compares no-skill, curated-skill, and self-generated-skill settings across diverse tasks, and its most useful finding for this thesis is that smaller and more focused skill sets can outperform larger and more comprehensive collections [9]. This weakens the monotonic assumption that adding more skills always helps: usefulness depends on how tightly skills are scoped, represented, and made selectable. The single-agent skill-failure paper studies scale more directly by formalising skills as descriptor-policy-backend tuples and increasing the number of available skills [8]. It finds that selection performance can remain stable until a threshold is reached, after which accuracy drops sharply. Crucially, that drop is linked not only to candidate count but also to semantic confusability among competing skills.

SkillRouter, SkillRet, and Skill Retrieval Augmentation make the same pressure concrete at large-repository scale, where an agent must identify relevant skills before downstream planning or execution [2, 19, 23]. SkillRouter treats routing as a large-scale retrieve-and-rerank problem over tens of thousands of candidate skills and shows that names and descriptions can under-specify the information required for correct selection [23]. SkillRet introduces a large-scale retrieval benchmark with public skills, structured semantic tags, a two-level taxonomy, training data, and thousands of evaluation queries [2]. SRA formulates a broader pipeline in which skills are retrieved, incorporated, and applied from large external corpora [19]. These papers establish skill retrieval as a standalone problem rather than a small prompt-engineering detail.

Scale and semantic confusability are related but not identical. Scale is the pressure created by many available skills and by the cost of exposing, embedding, ranking, or reranking them. Semantic confusability is the pressure created when several skills look relevant to the same request even though only one is procedurally correct. A large library makes confusability more likely because it increases the chance that multiple skills occupy the same domain, but a smaller domain-specific library can still be difficult if candidates share the same surface vocabulary. A library with many unrelated skills is mostly a search and context-management problem. A library with several similar PDF, spreadsheet, code-review, or deployment skills is a procedural disambiguation problem. This thesis is primarily concerned with the second failure mode: the wrong skill should often be plausible by topic, while the gold skill should be better because of an operational distinction.

2.3 Operational Information in Skill Artifacts

If semantic confusability is the core failure mode, the next question is what information might separate plausible near-neighbour skills. The field groups considered in this thesis should be understood as *candidate operational information*, not as a claim that the thesis invents a universal skill schema or already knows which fields will work best. They are proposed axes along which two skills can be procedurally distinct: use condition, input or precondition, output artifact, workflow or procedure, dependency or resource requirement, boundary or not-for condition, and success or verification evidence. A difference on one of these axes matters only when it would change correct invocation, feasibility, execution path, output, or verification.

The conceptual skill literature motivates this operational view. Procedural-memory and SoK-style accounts imply that selecting a skill requires more than matching a topic: the selector must know whether the procedure applies, whether the expected inputs and resources are available, what actions the skill will perform, and what result counts as successful [5, 21]. These are not merely formatting preferences inside a skill document. They are possible reasons why one skill should be invoked and another similar skill should not.

The Scheduling-Structural-Logical (SSL) representation provides a clear recent skill-specific schema example [10]. SSL separates skill information into a scheduling layer, a structural layer, and a logical layer. Its scheduling layer includes signals such as supported intents, expected inputs, expected outputs, dependencies, and control-flow features. Its structural layer captures scene-level execution phases, while its

logical layer records actions, resources, conditions, and effects. These categories overlap strongly with the thesis field groups. Use condition, input or precondition, output artifact, dependency or resource, workflow or procedure, and success or effect evidence should therefore be read as shared representational territory with SSL-like work, not as isolated labels invented here. The unresolved question is more specific: which of these information types are actually discriminative for routing when skills are deliberately near neighbours?

Public skill systems show why these signals are plausible but difficult to interpret. Anthropic-style filesystem skills may package instructions, scripts, resources, references, dependency instructions, and examples [1]. The Claude-skills ecosystem analysis shows that public skill libraries are uneven in length, redundancy, authoring style, and supply-demand balance [12]. Real artifacts may therefore contain routing-relevant information, but it may be implicit, redundant, inconsistently named, or mixed with execution-support material. A use condition may appear as a trigger phrase, a warning, or a sentence inside the workflow. A dependency may be a hard routing constraint, a setup command, or merely an implementation note. A boundary may be explicit, such as a warning not to use a skill for regulated advice, or implicit, such as a skill only being safe for local files.

This matters because not every part of a skill artifact is equally useful for selection. Few-shot examples, sample commands, setup snippets, and reference links can help the agent use a selected skill, yet they do not necessarily define a different capability. Other information is closer to the invocation decision because it describes when a procedure applies, what it expects, what it produces, what it depends on, where it should not be used, or how success is recognised. The literature therefore motivates RQ1 without answering it. Existing work shows that skills are procedural artifacts, that structured schemas such as SSL expose operational information, and that public skill libraries contain such information unevenly. It does not yet determine which information types help distinguish semantically similar but procedurally distinct skills.

2.4 Skill Representation and Retrieval Strategies

The next literature question is how skill information is made usable for selection. This requires separating three objects that are often blurred together. A *skill artifact* is the stored capability itself: a

`SKILL.md` file, script, resource folder, generated program, prompt module, or other executable procedural package. A *skill representation* is the selector-visible form of that artifact: a name, short description, full text body, structured field card, hierarchy node, or graph neighbourhood. A *skill retrieval strategy* is the method that operates over those representations to choose candidates, such as flat lexical or dense matching over a text encoding, reranking a first-stage candidate set, traversing a hierarchy, or exploring a graph. This distinction matters because an experiment can improve selection by changing the artifact, by exposing a richer representation, or by using a stronger retrieval strategy.

Flat or unstructured retrieval is the first family. In this family, the selector sees a text encoding of the skill rather than an explicitly separated procedural structure. That encoding may be a short name–description pair, a longer metadata block, the full skill body, or a concatenation of selected fields. Lexical search, dense embedding retrieval, and simple hybrid retrieval are therefore matching variants inside the same family, not separate representation families: all rely on an unstructured or weakly structured text encoding. SkillAct is best read as a flat textual skill-representation baseline because skills are inserted into the agent prompt rather than retrieved through a learned index [13]. Voyager stores executable programs as skill artifacts, but its retrieval-visible representation is compact: each program is indexed through a generated natural-language description and recalled by description-level similarity when a new task is attempted [20]. These examples show why compact metadata is attractive. It is easy to author, easy to expose, and compatible with standard retrieval pipelines. They also expose the central uncertainty. A short description may be efficient but under-specify operational distinctions, while a full skill body may expose more information but increase selector-visible context, embedding cost, and retrieval noise.

Retrieve-then-rerank should be treated as a pipeline variation over this candidate set unless the reranker uses a different structured representation. SkillRouter, SkillRet, and SRA make this large-scale setting explicit by treating skill selection as retrieval, or retrieval followed by reranking, over large candidate pools [2, 19, 23]. A retrieve-then-rerank system first obtains a smaller candidate set using a cheaper flat matcher, then uses a stronger model or richer matching function to reorder candidates with more context. ToolRerank provides transferable evidence from tool retrieval that hierarchy-aware reranking can improve selection and downstream execution, while ToolRet cautions that generic retrieval models are not automatically well aligned with tool-like capability selection [18, 24]. For skill retrieval, reranking is promising because it can inspect a smaller set of plausible candidates after a cheaper first-stage retriever. However, reranking does not remove the representation question. If the first-stage retriever fails

to include the gold skill, or if the candidate representation hides the operational distinction, a reranker may not recover the correct skill.

Hierarchy, tree routing, and DAG-style orchestration sit between representation and retrieval. They can define what the selector sees, but they also define how candidates are traversed or filtered. Adjacent retrieval and tool-use work supports the general coarse-to-fine intuition. RAPTOR organises document chunks into a recursive summary tree and retrieves across abstraction levels rather than relying only on flat chunks [16]. ToolChain* is closer to tool use: it represents possible API-call sequences as a decision tree and uses A* search to prune the action space, reducing exhaustive exploration while preserving a path toward a valid tool plan [26]. Skill-specific work makes a similar claim in the skill-retrieval setting. The single-agent skill-failure paper and AgentSkillOS suggest that coarse-to-fine selection can mitigate failures that appear when an agent chooses directly from a large flat pool [7, 8]. The attraction is clear: hierarchy can reduce search space, context cost, and the number of decisions made at once. The risk is also clear. A hierarchy assumes that early routing choices are reliable and that the gold skill is reachable through the chosen branch. Real skill libraries may violate this assumption because skills overlap across domains, combine capabilities, or differ procedurally inside the same topical category.

Graph and relational approaches generalise this issue. RAG and tool-use work provide adjacent methodological support here. KG²RAG starts from semantic retrieval, then expands and organises chunks through knowledge-graph relationships, showing how graph links can add structure after a flat first-stage retriever [25]. SciToolAgent provides a closer capability-selection example, where prerequisites, dependencies, compatibility, and safety constraints are represented in a graph for scientific tool use [3]. Skill-specific graph work applies the same idea directly to skills: SkillNet models skills through taxonomy, typed relations, and package structure, while knowledge-graph work on learned skills suggests that skill procedures can be decomposed into structured relational forms [11, 22]. Graphs are attractive because they can represent multi-parent membership, similarity, dependency, composition, resource use, and workflow compatibility without forcing a single tree path. However, graphs are only as useful as their nodes and edges. If graph edges are built from shallow semantic similarity, the graph may reproduce the same ambiguity that confused flat retrieval. If edges encode procedural relations, dependencies, boundaries, or outputs, the graph may support more targeted selection.

Structured field representations provide another representation choice. SSL is one clear recent example of converting text-heavy skill artifacts into linked scheduling, structural, and logical layers [10]. This approach is relevant because it makes operational information more explicit before retrieval. A structured

representation can in principle preserve inputs, outputs, dependencies, workflow phases, resources, conditions, and effects more compactly than a raw full document. Yet it also introduces new uncertainties. The schema must be extracted or authored reliably; the retriever must know how to use the fields; and the evaluation must distinguish the value of the structure from the value of simply adding more text. Structured representations therefore provide a plausible answer to the representation problem, but the literature has not yet fully separated field quality, extraction quality, retrieval strategy, and evaluation design.

The literature therefore motivates several RQ2 candidates rather than a single settled architecture. Plausible skill representations include unstructured text encodings, such as flat metadata or full-body text, hierarchy nodes, graph relations, and structured fields. Plausible retrieval strategies include flat lexical or dense matching over text encodings, retrieve-then-rerank pipelines, hierarchy traversal, graph traversal, field-aware matching, and hybrid pipelines. These choices differ in what information they expose, how much context they require, where they place the main selection decision, and what failure modes they create. The unresolved comparison is how different representation and retrieval strategies preserve, organise, or exploit the operational information introduced by RQ1.

2.5 Public Skill Ecosystems and Noisy Authored Skills

Public skill ecosystems add an important practical constraint. The data-driven analysis of Claude skills contributes ecosystem evidence rather than a new retrieval method [12]. It measures properties of a large public skill collection, including redundancy, length variation, and supply-demand imbalance. This is relevant because it shows that real skill libraries are already noisy, redundant, and unevenly structured. Some skills are narrow and atomic. Others are broad wrappers, parent skills, documentation-heavy packages, or lightly modified variants of similar capabilities. Such variation makes retrieval both a scale problem and a representation-quality problem.

Public skills also show why extraction and normalisation may be needed. Real artifacts rarely contain only a name and one-sentence description. They often include use conditions, expected inputs, workflow instructions, output expectations, dependency notes, resource files, examples, setup commands, and warnings about when not to use the skill [1, 12]. These elements can be candidate selection evidence, but they are not always cleanly labelled. For example, a dependency line such as an `npm install`

command may simply support execution after a skill is selected, while a dependency on a specific platform, package version, account, or runtime may be a hard routing constraint. Similarly, a boundary warning may be decisive if it excludes legal, financial, medical, or destructive use, but weaker if it is a generic caution repeated across many skills.

This public-skill evidence supports the thesis decision to separate controlled near-neighbour experiments from public-gold and external-scale checks. The controlled setting isolates semantic confusability and field-level distinctions: non-target information can be held constant while one operational field is exposed or hidden. Public settings test whether the same ideas survive messier authored artifacts, broad parent skills, duplicate capabilities, and provider-specific names. The external SkillRouter-style benchmarks add further portability pressure, but they can differ from the local controlled setting in important ways, including prompt length, explicit file references, and how normalised the skill bodies already are. Public corpora therefore help test ecological validity, but they do not replace controlled field-isolation tests for RQ1.

2.6 Evaluation Gaps

One of the clearest gaps in the literature emerges when evaluation practices are examined closely. Existing work often reports overall task success, while fewer studies isolate retrieval quality itself. Skills-Bench measures whether skills help, but not large-library disambiguation [9]. AgentSkillOS and SkillNet demonstrate strong system-level gains, but retrieval, orchestration, and planning improvements are partly entangled [7, 11]. The Claude-skills analysis documents redundancy, length variation, and ecosystem unevenness, but does not compare retrieval methods experimentally [12]. The single-agent skill-failure paper directly targets semantic confusability, but it studies synthetic skill descriptors and a relatively simple hierarchy rather than field-level evidence from authored skill artifacts [8]. Adjacent tool, document, and RAG retrieval work supplies useful ideas, but does not directly evaluate procedural skill artifacts [3, 16, 18, 24–26].

Taken together, these works leave three gaps that map onto the two research questions. The first gap is field-level and motivates RQ1: existing work shows that skill bodies and structured skill representations can matter, but it does not isolate which operational information types carry discriminative routing value under semantic confusability. A benchmark may show that full skill text helps, or that a structured representation improves a task, without showing whether the decisive signal came from input requirements,

output expectations, workflow steps, dependencies, boundaries, success criteria, or some other part of the artifact. The second gap is architectural and motivates RQ2: retrieval quality, representation design, reranking, orchestration, and downstream task success are often entangled. Some work begins to separate these stages, but system-level gains still do not always reveal whether improvement came from a better skill representation, a stronger first-stage retriever, a better reranker, or better use of the selected skill. The third gap is cost-related and also belongs to RQ2: even when efficiency is reported, cost is not consistently decomposed into comparable selector-visible tokens, candidate-budget size, retrieval latency, reranking cost, and amortised extraction or indexing cost.

These limitations mean that the thesis evaluation must separate several questions that prior work often combines. It must ask whether an operational field is discriminative at all; whether a representation exposes that field compactly and accurately; whether the first-stage retriever includes the gold skill in the candidate set; whether a reranker or field-aware matcher can order plausible candidates correctly; and what cost is paid at indexing time and at query time. This is why RQ1 and RQ2 are treated as equal rather than sequentially optional. RQ1 identifies which information is worth preserving for near-neighbour distinction. RQ2 asks which representation and retrieval strategies preserve, organise, or exploit that information reliably and efficiently.

2.7 Synthesis and Research Questions

The literature reviewed above points to a common problem but does not close it. Skills are reusable procedural capability units, not merely documents or atomic tools. Large libraries create both scale pressure and semantic confusability. Public skill artifacts contain operational information, but it is unevenly expressed and mixed with execution-support material. Structured representations, hierarchies, graphs, dense retrieval, lexical retrieval, and retrieve-then-rerank pipelines are all plausible responses, yet the literature does not clearly determine which operational information most effectively distinguishes competing skills or which representation and retrieval strategies use that information best.

This motivates the two revised research questions of the thesis:

RQ1: Which operational information types help distinguish semantically similar but procedurally distinct agent skills?

RQ2: How do skill representation and retrieval-pipeline choices affect the preservation and use of the routing-relevant operational information identified in RQ1 under semantic confusability, and what accuracy, candidate-recall, and retrieval-cost trade-offs result?

These questions arise from the literature rather than being imposed on it. Existing work has already shown that skills are useful, that flat selection becomes unreliable under scale, and that richer organisation may help. What remains unresolved is which information actually carries discriminative routing value, and how different retrieval strategies preserve or lose that value. The rest of the thesis therefore evaluates skill retrieval under semantic confusability: not merely whether a selected skill is broadly relevant, but whether it is the procedurally correct capability among plausible alternatives.

Conceptual Framework

Chapter 2 establishes the literature-facing definitions of skill artifacts, skill representations, skill retrieval, semantic confusability, and candidate operational information. This chapter keeps only the experiment-facing frame used to connect those definitions to the benchmark and methodology.

3.1 Representation Conditions

The benchmark exports several representation conditions:

- R1 flat metadata.:** Skill name, family/category, and short description. This approximates the compact view used by progressive disclosure systems.
- R2 extracted procedural card.:** Use conditions, input/precondition fields, output fields, workflow/procedure fields, and boundaries. This tests which operational information improves retrieval under semantic confusability.
- R3 dependency/resource-aware card.:** R2 plus tools, external dependencies, resources, platform assumptions, and public-source metadata. These fields are useful for feasibility and context loading, but can introduce noise if treated as ordinary positive text.
- R4 graph edges.:** A planned heterogeneous graph view where skills may connect to typed nodes such as triggers, outputs, workflow steps, preconditions, dependencies, resources, provenance, or avoid conditions. The final thesis-facing edge schema is still to be decided.
- Full artifact.:** The complete `SKILL.md` text. This is important as a strong dense-retrieval baseline because it exposes the most information, but it is costly and can be noisy.

The R2/R3 conditions should be read as compact diagnostic carriers for operational information, not as a final universal skill-interchange standard. This makes them useful for controlled experiments because individual fields can be hidden, exposed, or ablated. A richer graph view remains relevant to RQ2, where

the question becomes how to organise and exploit the same information, rather than whether the field itself contains discriminative signal.

3.2 Retrieval Architecture Families

The architecture families are separated from the representation conditions:

M0 progressive disclosure.: The main agent sees compact skill cards and decides which full skill documents to load.

M1 flat lexical retrieval.: BM25 or TF-IDF over R1. This is a lexical control, not the proposed architecture.

M2 dense retrieval.: Embedding retrieval over R1, R2, or full artifacts.

M3 structured-card retrieval.: Retrieval over serialised R2 procedural fields.

M4 tree routing.: Hierarchical selection through family or cluster nodes before selecting a skill.

M5 graph retrieval.: Retrieval or reranking over R4 relation edges.

M6 hybrid procedural reranking.: A first-stage retriever produces a candidate set, then a procedural layer reranks candidates.

M7/M8 provider retrieval and reranking.: Modern API embedding and reranking baselines, currently using Qwen embeddings and Qwen reranking.

The current deterministic schema reranker is treated as **M6-v0**. It is useful as a transparent diagnostic but is not the final proposed procedural reranking method. The planned **M6-v1** method should parse the user request into fields and compare those request fields against extracted skill fields. This distinction prevents the thesis from overclaiming that simple text overlap is equivalent to structured reasoning.

3.3 Tree and Graph Structure

Tree and graph methods are experimental architectures and information carriers, not automatically the same information condition as R2/R3 fields. M4 tree routing tests whether a hierarchy can reduce candidate set size without excluding the gold skill too early. M5 graph retrieval tests whether explicit typed relations improve retrieval beyond serialised structured cards. The exact branch summaries and edge types are still planned and should be treated as TBD until evaluated.

The current experimental R4 export includes candidate edge types such as:

- `belongs_to_family;`
- `triggered_by;`
- `avoid_when;`
- `has_precondition;`
- `has_workflow_step;`
- `produces_output;`
- `requires_dependency;`
- `has_resource_signal;`
- `uses_resource;`
- `derived_from_public_skill.`

Some of these edges can be derived from the same operational information used in R2/R3, but graph retrieval should not be described as using "the same information" unless the mapping is explicitly implemented. Other edges may carry relation, provenance, grouping, dependency, or similarity signals that are better understood as RQ2 strategy information. This allows later edge ablations: for example, testing trigger-only retrieval, trigger plus output, trigger plus output plus workflow, and then adding dependencies, resources, and avoid penalties.

The graph should not include benchmark-only edges such as `confusable_with` if those edges are derived from gold labels, closest alternatives, or manual adjudication. Such edges are useful for analysis, but using them for retrieval would leak the benchmark structure. Final M5 retrieval should use only skill-artifact information, public provenance, or unsupervised similarity that would be available outside the evaluation set.

3.4 Working Expectations To Test

The experiments are guided by five working expectations. These are not assumed conclusions; they define what the benchmark is designed to test.

- (1) Flat metadata may under-specify procedural distinctions under scale, but it can remain competitive when names and descriptions expose sufficient task or category cues.

- (2) Full skill artifacts will improve over short descriptions in some cases because they expose more information, but they may also introduce noise and cost.
- (3) Input/precondition, use condition, and output/artifact fields are expected to be strong positive fields for distinguishing near-neighbour skills.
- (4) Dependency/resource, boundary/not-for, success/verification, and workflow/procedure fields are expected to be more conditional because they often require capability context, authority inference, acceptance-gate reasoning, or procedure-order reasoning.
- (5) Hybrid retrieval-reranking methods may improve final order only when first-stage retrieval retains the strict gold within the reranker candidate set.

The completed RQ2b result evaluates these as questions rather than assumptions. In particular, it separates the observed accuracy of I1, I2, I3-flat, and I3C from first-stage candidate inclusion, fixed-list reranking, and measured selector cost. It does not evaluate a tree or graph architecture, so those remain future architecture hypotheses rather than competing current result rows.

Benchmark Design

4.1 Design Goals

The benchmark is designed to test skill retrieval under the failure mode that motivates the thesis: semantic confusability between procedurally distinct skills. It is not enough for the benchmark to contain many unrelated skills. The benchmark must contain clusters where several skills sound relevant to the same request, but only one skill is procedurally correct.

The benchmark therefore has four goals:

- (1) create controlled near-neighbour skill clusters;
- (2) scale the library with generated and public background skills;
- (3) validate that gold labels remain defensible under non-core competition;
- (4) test whether proposed representation fields are also visible or extractable in public skill artifacts.

4.2 Skill Library Composition

The frozen-v0.4 benchmark contains 2433 skills:

- 1800 generated background-scale skills;
- 460 imported public skills;
- controlled confusable skill clusters across multiple task families;
- 32 public-style controlled skills;
- 10 implicit-field stress skills;
- support skills used for older messaging and email cases.

The controlled evaluated skills are organised into task families such as document and PDF operations, browser automation, GitHub and CI workflows, Hugging Face workflows, observability, office automation, API design, deployment, planning, reading and research, messaging, security, and skill lifecycle operations. These skills form the main controlled benchmark. The public skills are used both as realistic background distractors and as evidence for whether the proposed fields exist or are extractable in real skill artifacts.

4.3 RQ2b V3 Strict-Gold Freeze

The thesis-facing full-library evaluation uses the later source-grounded V3 freeze rather than the historical frozen-v0.4 endpoint. It contains the same scale of 2,433 skill artifacts, but its scored subset is deliberately narrower: 381 strict-gold prompts, comprising 243 controlled prompts and 138 public-gold prompts. Each scored prompt has one frozen intended gold skill. The metric is therefore agreement with this intended strict label; it is not a claim that no other skill could ever be useful for the request.

For each source skill, the V3 freeze retains the original source path and SHA-256, native name and description, original full body, and source-grounded I3 evidence. Automatic validation checked source identity, literal evidence provenance, heading exclusion, serialisation, duplicate IDs, and complete source coverage across 57 extraction chunks. A protocol-invalid manual semantic review attempt was quarantined and no replacement human-semantic completeness audit was used. The full-library results must consequently be read as source/provenance-validated strict-label retrieval evidence, not as a universal human judgement that every extraction is semantically complete.

4.4 Controlled Confusable Clusters

A valid controlled cluster must satisfy three conditions. First, the skills must overlap in surface meaning, name, description, or trigger phrases. Second, the skills must differ in operationally important procedure. Third, choosing the wrong skill must matter for the resulting workflow or artifact.

Examples include:

- **PDF/document operations:** PDF question answering, layout table extraction, OCR cleaning, form filling, redaction review, and PDF-to-DOCX conversion.

- **CI and GitHub workflows:** CI log root-cause debugging, PR review-comment resolution, repository code review, issue triage, changelog generation, and git safety guardrail installation.
- **Observability:** Prometheus alert writing, dashboard building, distributed trace investigation, SLO breach narrative writing, resilience review, and service-mesh traffic debugging.
- **Skill representation analysis:** skill field auditing, router policy design, skill deduplication, benchmark cluster construction, and graph-edge schema design.

The purpose of these clusters is not to model every possible agent skill. It is to create controlled pressure around the exact failure mode of interest: plausible semantic neighbours with different procedures.

4.5 RQ1 Field-Isolation Suites

The RQ1 field-isolation suites are a more controlled derivative of the same near-neighbour idea. They are not intended to be another broad benchmark version. Instead, each suite isolates one operational field while holding all other skill-side information constant. A suite contains 50 clusters, and each cluster contains three sibling skills: one intended gold skill and two deliberately plausible alternatives. The siblings share the same broad task family and the same non-target operational context. Only the target field differs.

The condition with the field hidden, `shared_context_only`, shows only the shared neutral context. In this condition, the sibling candidates are intentionally tied. The condition with the field shown, `shared_context_plus_field`, shows the same shared context plus one target-field line. This design lets RQ1 ask whether the field itself carries useful information for routing. It also separates RQ1 from the broader RQ2 question of which representation or retriever is best.

The completed RQ1 suites cover use condition, input/precondition, output/artifact, dependency/resource, boundary/not-for, success/verification, and workflow/procedure. The examples/tests suite is retained as a negative control, because examples can steer selection under exact scenario matches but do not necessarily define a distinct skill capability. The units were created with AI assistance under a researcher-defined rubric, mechanically checked, and reviewed and approved by the researcher. The labels were visible during this review, so it was not an independent blinded review.

4.6 RQ1 Public Original-Document Benchmark

The second RQ1 experiment uses preserved public skill documents rather than controlled field templates. A source-tracked pool of 1,099 originals supplies 82 candidate groups containing 265 documents. Each group contains natural near-neighbours and direct/paraphrased requests with one frozen gold skill. Candidate grouping and label preparation used source evidence and AI assistance. The researcher then reviewed and approved all 194 scored gold-label cases with the labels visible. This records researcher agreement, not independent blinded annotation.

The preparation pipeline produced 574 possible candidate-group and field removals. Of these, 439 passed the technical checks for every candidate, and 402 also had an eligible request and gold skill and therefore entered scoring. The upstream registry contains 199 request/gold families, but only 194 occur in at least one scored field condition. Reusing eligible families across fields produces 1,078 field-specific cases and 2,156 direct/paraphrased request rows. These numbers describe different stages of preparation and must not be treated as counts of independent candidate groups.

For each scored case, the intact public documents form the `FULL_ORIGINAL` condition. The paired condition keeps the same prompt, candidates, order, source hashes, and gold skill, but removes every cited source line assigned to the target field from every candidate. The implementation replaces only those line contents with blanks and leaves the remaining text unchanged. Each field has its own eligible case count because public documents do not expose equally removable evidence for every field. An edited document may be incomplete or non-executable; this benchmark tests routing after information removal, not execution of the edited skill. Three supporting tests remove predefined groups of fields. They measure the combined removal and do not assign the result to one member field or assume that individual effects add together.

4.7 Background and Public Skills

The 1800 generated background-scale skills provide scale pressure and near-domain distractors. They are designed to be functional enough to compete semantically with evaluated prompts, but they are not the main gold-label targets. Their role is to reveal whether retrieval methods are robust when the library contains many plausible non-core candidates.

The 460 public imported skills improve ecological validity within the historical frozen-v0.4 library. They introduce authoring styles, broad skills, implicit fields, resources, dependencies, and hierarchical patterns that are less clean than the controlled skills. Those embedded public skills are used in three ways:

- (1) as background scale and distractors;
- (2) as a field-taxonomy audit corpus;
- (3) as a separate public-gold validation stratum when a public skill can be manually adjudicated as a stable target.

The 460-skill corpus is not the current prevalence basis for the RQ1 field taxonomy. A later, separate source frame freezes 29,292 canonical public `SKILL.original.md` documents. The review was completed in 31 batches by a Codex `gpt-5.6-terra` agent using `xhigh` reasoning. For each document, the agent read the `YAML` `name` and `description` metadata together with all Markdown sections, tables, and examples, but not attached scripts, references, or other auxiliary files. It recorded only whether each candidate information type was present. The retained package contains the aggregate summary and batch totals, but not the 29,292 row-level decisions. This review asks whether the candidate fields recur in naturally authored public skill documents; it does not make the skills gold targets, certify document quality, or test a retrieval effect.

4.8 Historical Frozen-v0.4 Public-Gold Validation Stratum

A separate 144-prompt public-gold stratum was built for the historical frozen-v0.4 benchmark to test whether externally authored public skills can serve as retrieval targets. This is deliberately separated from the controlled benchmark because public skills are often broader, less atomic, duplicated, or hierarchical.

At that historical endpoint, the public-gold stratum was scored with both strict gold labels and documented acceptable alternatives. Its validation status was:

- 144/144 prompts pass integrity;
- 144/144 prompts pass procedural distinctness;
- 131/144 prompts pass prompt-specific alignment;
- 144/144 prompts pass semantic confusability;

- 0 critical exact-name leaks and 0 high-risk leaks.

These historical validation results support a cautious claim: public skills are valuable for external validity, but they are not clean gold labels by default. They must not be conflated with the thesis-facing V3 RQ2b endpoint, which contains 138 source-grounded public-gold prompts within its 381-prompt strict-only overlay and does not report acceptable-alternative scoring.

4.9 Gold Labels and Acceptable Alternatives

Each evaluated prompt has a gold-label intended skill. A gold label is the skill that should be selected if the selector correctly interprets the user’s procedural requirements, not merely the skill whose name or description is most semantically similar to the prompt. A gold label is valid when careful reading shows that the prompt’s requirements align more strongly with the gold skill than with its near neighbours. Historical frozen-v0.4 records also retain documented acceptable alternatives; V3 RQ2b reports strict intended-label agreement only and does not calculate, imply, or substitute an acceptable-at- k endpoint.

Operationally, a skill is treated as the gold label only when it satisfies the following criteria:

- (1) **Procedural fit:** the skill’s intended use, inputs or preconditions, workflow, output artifact, and success criterion match the request.
- (2) **Discriminative fit:** at least one concrete requirement in the prompt explains why this skill is better than the listed alternatives.
- (3) **Near-neighbour pressure:** the alternatives are semantically plausible, but differ from the gold skill on an operational axis such as input state, output type, workflow, dependency, boundary condition, or success criterion.
- (4) **Atomicity:** the skill is specific enough to be selected as a single procedural capability. Broad router skills or hierarchical skills are not used as strict gold labels unless the prompt explicitly asks for that broad routing behaviour.
- (5) **No leakage:** the prompt should not reveal the gold skill through exact skill names, copied distinctive phrases, or artificial labels.
- (6) **Stability under scale:** when public and generated background skills are added, no unrecorded competitor should be clearly more procedurally suitable than the gold skill.

These criteria distinguish the gold label from a merely plausible retrieval hit. For example, in a PDF cluster, a request to extract native text, tables, and metadata from a non-scanned PDF has a different gold skill from a request to OCR a scanned PDF, fill a PDF form, answer questions over a PDF, or convert a PDF to DOCX. All of those alternatives are semantically close because they concern PDFs, but they are procedurally different.

Acceptable alternatives are recorded when a competing skill is genuinely near-equivalent. This avoids treating annotation ambiguity as either a retrieval success or retrieval failure. In public-gold cases, acceptable alternatives are especially important because imported public skills can overlap with controlled skills or with other public skills.

4.10 Validation Rubric

The benchmark is validated through a staged rubric:

- (1) **Integrity:** all prompts, gold skills, and alternatives resolve.
- (2) **Procedural alignment:** each prompt contains requirements that justify the gold skill.
- (3) **Semantic confusability:** each prompt has plausible near-neighbour skills.
- (4) **Prompt leakage:** prompts should not reveal exact skill names or copied distinctive phrases.
- (5) **Scale regime:** the library should be large enough to create context and retrieval pressure.
- (6) **Public field audit:** proposed fields should recur in a frozen public primary-document corpus, or be clearly labelled as proposed normalisation fields.
- (7) **Public-gold validation:** public skills should be manually atomised and adjudicated before being used as gold targets.
- (8) **Selector pressure:** baseline methods should show meaningful performance differences rather than all succeeding or all failing randomly.
- (9) **Downstream validation:** a smaller sample should test whether retrieval quality affects task success.

4.11 Historical Frozen-v0.4 Validation Status

The frozen-v0.4 validation status is:

- Step 1 integrity: passed for 245 controlled prompts, 144 public-gold prompts, and 2433 skills.
- Step 2 procedural distinctness: controlled 245/245 prompts and 830/830 pairs pass; public-gold 144/144 prompts and 575/575 pairs pass.
- Step 2 prompt-specific alignment: controlled 238/245 prompts pass; public-gold 131/144 prompts pass.
- Step 3 semantic confusability: controlled 243/245 prompts pass; public-gold 144/144 prompts pass.
- Step 4 prompt leakage: passed with 0 critical exact-name leaks and 0 high-risk leaks in both controlled and public-gold strata.
- Step 5 scale regime: passed; R1 flat metadata is approximately 333k selector-visible tokens, while full skill artifacts exceed 1M selector-visible tokens.
- Step 6 public field audit: the historical frozen-v0.4 record audited 460/460 imported public skills heuristically and with DeepSeek model-assisted verification; the later 29,292-document full-primary-document census is the current descriptive prevalence evidence.
- Step 6b public-gold validation: current public-gold stratum passes validation but should still be reported separately from controlled results.

The historical frozen-v0.4 validation record remains useful for provenance, but it is not the scoring endpoint for RQ2b. The RQ2b V3 strict-gold freeze uses the narrower 381-prompt subset described above and reports controlled and public-gold strata separately.

Methodology

5.1 Experimental Overview

The methodology is a controlled comparison of operational-information, representation, and retrieval conditions. Each method receives the same user prompts and skill library and produces a ranked list of candidate skills. Evaluation follows the label regime defined for the relevant experiment: historical frozen-v0.4 records retain documented acceptable alternatives, whereas V3 RQ2b reports strict intended-label agreement only.

The methodology separates two related questions. RQ1 is an information-isolation question: which operational fields help distinguish semantically similar skills when skills, prompts, and retrieval setup are otherwise held fixed? RQ2 asks how skill representation and retrieval-pipeline choices affect the preservation and use of those fields under semantic confusability, and what accuracy, candidate-recall, and retrieval-cost trade-offs result. This separation is essential because a selector can fail either because the visible representation has lost the relevant operational distinction, or because the selection method fails to use information that is present.

This methodology does not claim that the list of operational fields is itself novel. It overlaps with SSL-style structured skill information: expected inputs and outputs, dependencies, execution phases, action/resource evidence, effects, and boundaries. RQ1’s contribution is a controlled way to test each field in near-neighbour routing, while prior structured representations mainly test the complete representation as a whole.

The independent variables are therefore:

- the operational information exposed to the retriever;
- the skill representation that carries or organises that information;

- the first-stage retrieval strategy;
- whether a reranker is used;
- whether the reranker uses selection information as structured fields or as ordinary text.

For RQ1, the main dependent measures are top-1 accuracy, mean reciprocal rank (MRR), lift over the hidden-field baseline, prompt-variant breakdown, and ambiguity resolution for reasoning selectors. Top-k recall, candidate budget, context cost, latency, and qualitative failure type mainly support RQ2 because they describe complete retrieval strategies rather than individual field discriminability.

5.2 RQ1 Methodology: Field-Isolation Test Units

RQ1 is evaluated before the broader representation/retrieval matrix. The purpose is not to ask which complete selector is best. The purpose is to test whether a specific skill-side operational field helps distinguish semantically similar sibling skills when everything else is controlled.

The first RQ1 experiment uses field-isolation test units. Each unit contains one prompt, one intended gold skill, and two deliberately plausible sibling alternatives. The siblings share all non-target context: broad task intent, the non-tested operational fields, workflow, boundary, dependency, and success framing. Only one target field differs. Each cluster has one direct prompt and one paraphrase-safe prompt. Boundary/not-for units additionally include an `implicit_authority` prompt variant, where the user asks for a constrained review role, triage note, or evidence packet without explicitly naming the excluded action. Dependency/resource units use a `contextual` prompt variant instead of ordinary paraphrase: the task request remains natural, while routing context supplies a positive capability cue such as a package manifest, available MCP server, credential, runtime, or provider configuration. This separates explicit user wording from capability evidence that may be acquired from the environment.

The main controlled comparison uses the same shared context in two conditions: first with the target field hidden, and then with one candidate-specific target-field line shown. In the experiment files, these conditions are called `shared_context_only` and `shared_context_plus_field`. The added line can be, for example, `Input / Precondition: ...` or `Output Artifact: ....`. Because the candidates are intentionally identical when the field is hidden, top-1 and MRR are reported with tie-aware scoring. Other runner conditions are diagnostics only: `field_only` asks whether the field text alone is enough signal, and `full_skill_doc` asks whether the signal survives when

embedded inside the whole generated skill document. These diagnostics are not the main RQ1 field-isolation comparison. The candidate-specific field values are used as written rather than forced to equal length. The result therefore measures the practical value of showing that information, including its amount and specificity; it does not separate field meaning from text length alone.

The key RQ1 reporting rule is therefore simple: a field is treated as useful only when exposing that field improves selection over the same prompts, the same sibling set, the same retriever, and the same scoring metric. The completed use-condition, input/precondition, output/artifact, dependency/resource, boundary/not-for, success/verification, and workflow/procedure suites establish this pattern. Prompt variants are averaged within cluster and uncertainty is estimated by 10,000 cluster bootstrap resamples, not by treating the variants as independent tasks. A companion leading-negative diagnostic subtracts the highest-scoring non-gold sibling's score from the gold score. It is a retrieval-leading-negative margin, not a human semantic-distance annotation, because the frozen units do not pre-designate one of the two alternatives as semantically closer. Use condition is reported with direct and paraphrase rows separated because it can act as a direct task-intent hint when the prompt names the same trigger, but may require stronger semantic interpretation when the request is paraphrased. Boundary/not-for is also reported with its prompt variants separated, because explicit boundary wording and implicit authority cues test different routing conditions. Dependency/resource is reported as capability compatibility rather than raw setup text: a positive dependency signal is a required tool, server, provider/API, credential boundary, runtime/platform, pinned version, database, hardware, config, or resource. Generic commands such as `npm install`, `pip install`, `npm ci`, `build/test/lint` steps, or dependency-hygiene advice are not counted as strong routing evidence unless they name a concrete capability or compatibility conflict. In the public-skill audit, for example, `npm install` and `npm ci` are weak setup evidence by themselves, while requirements such as a Chrome DevTools MCP server, `GH_TOKEN`, or Node.js 20+ are candidate routing evidence because they constrain which skill can run in the current environment. Success/verification is reported only for concrete acceptance gates, such as schema validation, threshold checks, evidence anchoring, unresolved finding state, rollback proof, reproducible failure, or smoke checks. Workflow/procedure is reported as procedure-order or operation-path evidence, with aspect types such as wrong order, swapped step, missing prerequisite, missing branch, missing verification, and substitute method. The examples/tests suite is reported separately as a negative-control/support-information suite, not as an eighth core routing field.

This is also the main difference from SSL-like representation work. SSL can show whether a complete structured skill record improves discovery overall. RQ1 asks the narrower question of whether one operational field supplies the missing distinction between otherwise equivalent sibling candidates.

The second RQ1 experiment tests the same question in preserved public skill documents. It starts from 1,099 source-tracked public originals; this is not a random sample and not every document was manually field-annotated. Source checking, near-neighbour screening, field-evidence mapping, prompt checks, and AI-assisted curation produced 82 purposefully selected candidate groups. For each accepted group and field, the selector compares the complete original documents with the same documents after every cited line carrying that field has been removed from every candidate. The implementation blanks those line contents so the line numbers remain stable; it does not add or rewrite any remaining text. The candidates, prompt, source hashes, and gold skill remain fixed. The edit may make a document incomplete or non-executable: the experiment tests routing after information removal, not execution of the edited skill. Each field has its own set of 49–69 eligible groups because a group is scored only when the target information can be removed cleanly from every candidate. Direct and paraphrased requests are averaged within each group, and 10,000 paired bootstrap samples show the uncertainty in `FULL_ORIGINAL - REMOVE_<FIELD>`. The removed passages differ in length, and one passage can express more than one operational function. The result therefore measures the effect of removing the identified information from these groups; it is not a length-controlled estimate of an isolated semantic category. The researcher reviewed and approved all 194 scored gold-label cases and all 402 scored removals with the labels or document edits visible; this was not an independent blinded review. Appendix B.7 shows one frozen controlled field-addition example and one frozen public line-removal example.

5.3 Current RQ2 Protocol Status

Both RQ1 experiments, the matched-content RQ2a fixed-candidate mechanism study, and the RQ2b full-library retrieve-then-rerank study are complete. RQ2a includes the protocol, deterministic cluster split, eight representation conditions, proposition/leakage validation, common result schema, selector adapters, length audit, development freeze, untouched confirmatory execution, paired statistical analysis, and cost ledger. The confirmatory split contains 280 clusters, 600 prompt variants, 22 conditions, and 13,200 aligned decisions. RQ2b scores 381 frozen strict-gold prompts against 2,433 skills, with 4,572 first-stage rows in total (1,524 per B1 retriever) and 9,144 fixed-Top-20 reranked rows in total

(4,572 per B2 reranker). Its result synthesis is explicitly post-hoc and descriptive; it reports all representation/retriever/reranker rows, paired cluster-bootstrap intervals, failure mechanisms, and measured cost without using a p-value or Holm threshold as a thesis claim gate. Earlier frozen-v0.4 I1/I2/I3, M6, and external SkillRouter experiments remain exploratory, feasibility, diagnostic, or portability evidence.

5.4 RQ2a Matched-Content Representation Study

RQ2a reuses all 350 author-reviewed controlled-RQ1 clusters and 750 prompt variants. Clusters, rather than individual prompts, are divided into a 70-cluster development split and a 280-cluster confirmatory split, balanced with 10 and 40 clusters per field respectively. The development split is used for templates, smoke tests, and the one permitted field-score aggregation decision. Confirmatory clusters must not be inspected for method tuning.

Each test unit keeps the query, three sibling candidates, gold identity, reviewed operational propositions, and deterministic candidate order fixed. It then changes only how the reviewed facts are exposed:

shared-only.: Shared neutral context with the decisive sibling-specific field values omitted.

same-facts-fielded.: All seven reviewed values with explicit field labels and boundaries.

same-facts-flat.: Exactly the same values and order as the fielded condition, but with labels and headings removed.

same-facts-prose.: The same propositions expressed through one frozen proposition-preserving prose template.

same-facts-diluted-2x.: The fielded values plus sibling-identical support text that contains no decisive cue.

Order-controlled and 0x/1x/2x/4x dilution variants are diagnostics after the core matrix. Every serializer must preserve the reviewed propositions, use neutral candidate names, omit benchmark role labels, record source and output hashes, and pass identity, leakage, proposition-equivalence, length, and padding checks.

The three core fixed-candidate selectors are BM25, Qwen `text-embedding-v4` single-vector cosine similarity, and direct cross-encoder scoring with `pipizhao/SkillRouter-Reranker-0.6B`. For each prompt, BM25 scores only the fixed three-sibling corpus, so its term statistics are local to that decision rather than global to the 280-cluster split. Qwen is a bi-encoder: the query and each complete

candidate representation are embedded independently, document vectors are computed once and reused, and cosine similarity produces the ranking. SkillRouter is a cross-encoder reranker model: each query–candidate pair is tokenised and scored jointly, allowing token-level interaction but preventing document-only precomputation of the final pair score. The cross-encoder scores all three candidates directly; in RQ2a it is therefore used as a selector, not as a reranker over a fallible first stage. This is distinct from the historical external pipeline that used the released SkillRouter embedding model for candidate retrieval before applying the reranker. One frozen LLM A/B/C/ABSTAIN selector is retained only as a reasoning diagnostic.

5.4.1 Semantic Field-Aware Selector

The repository already contains M6-v1 lexical and M6-v2 semantic field-aware prototypes. These establish implementation feasibility, but they do not satisfy the clean RQ2a protocol: they are tied to old representations, selected field sets, calibrated weights, request parsing, boundary penalties, or first-stage-score blending. The implemented RQ2a field-aware selector adapts reusable M6-v2 embedding and caching components rather than claiming a from-scratch method or reusing the old result rows.

The adapter reads `same-facts-fielded`, embeds the raw query, and embeds all seven candidate fields separately. It directly scores all three candidates without candidate exclusion or a first-stage prior. It does not receive the target field, cluster type, gold identity, candidate role, or alternative metadata. For every query-candidate pair it retains seven component similarities, an aggregate score, selected candidate, rank margin, and latency. Query and field embeddings are cached by exact text.

Only two target-agnostic aggregation rules were compared on the development split: maximum field similarity and the uniform average of the two highest field similarities. The uniform-top-two rule was frozen and then applied unchanged to every confirmatory field suite. Field-specific weights, oracle target-field scoring, and tuning on confirmatory clusters were prohibited. This creates two distinct mechanism contrasts:

- fielded versus flat under Qwen single-vector selection tests whether labels and boundaries help a generic semantic selector;
- field-aware versus single-vector Qwen on identical fielded text tests whether explicit field-wise use adds value beyond merely displaying the structure.

The primary RQ2a outcome is paired top-1 accuracy at the cluster level. Secondary outcomes are MRR, score margin, near-neighbour confusion, prompt-variant and field strata, selector-visible tokens, and latency. Each preregistered contrast uses 10,000 cluster bootstrap resamples and 10,000 paired cluster sign flips; the eight secondary contrasts receive full-family Holm correction. Before confirmatory execution, a three-percentage-point top-1 difference was frozen as the smallest effect of practical interest. A technically valid null result remains an answer: it may show that preserving facts matters more than field labels, or that explicit field segmentation does not help under the frozen aggregation rule.

5.5 RQ2b Full-Library Representation And Pipeline Study

RQ2b evaluates source-grounded representations over the V3 strict-gold library. Every condition uses the same 2,433 source skills and 381 prompts, separated into 243 controlled and 138 public-gold prompts. The labels are single intended gold skills frozen before scoring. They support strict-label retrieval comparisons, not a claim that each selected label is the uniquely useful skill in every real deployment.

The four representation conditions preserve a clear content boundary:

- I1.:** Native skill name plus frozen source-native description.
- I2.:** Complete original source skill body.
- I3-flat.:** Retained I3 source-evidence items serialised without field headings.
- I3C.:** The same retained I3 source-evidence items serialised with the seven operational field headings: use condition, input/precondition, output/artifact, workflow/procedure, success/verification, boundary/not-for, and dependency/resource.

I3-flat and I3C therefore contain the same extracted operational evidence; headings are the intended difference. I1 is not a “no-information” condition: it can contain name and description cues. I2 is not a matched-content control because it retains all body text, including information that I3 may omit. Canonical extraction records retain each source path and SHA-256 and validate the source grounding of every retained evidence item.

B1 independently ranks the full library with local BM25, Qwen `text-embedding-v4` bi-encoder retrieval, and the native SkillRouter bi-encoder. Qwen scores a query against the maximum cosine across any document chunks for a skill; its cached document vectors are one-time construction work. B2 takes the immutable B1 Top-20 list for each representation/retriever/prompt condition and reorders only

those candidates with Qwen qwen3-rerank or pipizhao/SkillRouter-Reranker-0.6B. Thus B2 tests conditional ordering quality, not recovery from a first-stage candidate miss. No result row changes the candidate identities, gold labels, sources, or representation text after B1.

The completed B3-v2 synthesis aggregates the frozen B1/B2 rows by semantic cluster and reports strict Hit@1, MRR@20, Recall@20, direct/paraphrase-averaged cluster effects, controlled and public-gold strata, paired cluster-bootstrap intervals, descriptive failure rates, and recorded cost. It contains no new model call or method tuning. B4 adds a provenance-bounded availability census: an I3 item is description-visible only when its original source text, after whitespace folding only, is present in the frozen I1 metadata description. This census is explanatory rather than causal; it cannot identify which text caused a ranking outcome.

5.6 Historical Exploratory Crossed Matrix

Table 5.1 records the earlier broad strategy matrix. It is retained for provenance and exploratory interpretation, not as the confirmatory RQ2a design. Its rows often change information content, representation, candidate generation, and reranking together, so they cannot isolate the new matched-content mechanism question.

Representation or carrier layer	Embedding only	Neural reranker	Field-aware reranker	Tree/graph retrieval
R1 flat metadata	Qwen and SkillRouter frozen-v0.4 run.	Qwen and SkillRouter frozen-v0.4 run.	Local and SkillRouter- first-stage top-20 runs complete.	Not central.
R2 structured procedural	Qwen and SkillRouter frozen-v0.4 run.	Qwen and SkillRouter frozen-v0.4 run.	Local and SkillRouter- first-stage top-20 runs complete.	Candidate ablation.
R3 dep./resource card	Frozen-v0.4 local ablations.	Not complete.	Dep. and boundary ablation.	Candidate graph ablation.
Full SKILL.md	Qwen and SkillRouter frozen-v0.4 run.	Qwen and SkillRouter frozen-v0.4 run.	Qwen and SkillRouter first-stage top-20 runs complete.	Requires field extraction first.

TABLE 5.1: Historical exploratory information-carrier and retrieval-architecture matrix. It is not the confirmatory matched-content RQ2a design.

This crossed design also explains why provider results must be interpreted carefully. For example, Qwen reranking over R2 structured cards is not directly comparable with SkillRouter reranking over full SKILL.md artifacts. That comparison changes both the model and the representation. The fair

comparison requires running each architecture against the same representation layer, especially R1, R2, and full skill text.

5.7 Information and Representation Conditions

The experiments use the following information and representation conditions:

R1 flat metadata.: Skill name, family, and short description. This approximates the compact representation that a main agent might see in a progressive-disclosure setup.

Full artifact text.: The complete `SKILL.md` text. This exposes the richest information but is expensive and can dilute the signal with examples, resources, and broad documentation.

R2 structured procedural card.: Extracted fields for use conditions, preconditions, workflow, outputs, and boundaries.

R3 dependency/resource-aware card.: R2 plus external dependencies, resource signals, public-source metadata, and bundled resource files.

R4 graph edges.: A planned typed edge view for relation-aware retrieval. Candidate edges may derive from operational fields, public provenance, dependencies, workflow order, input-output direction, or unsupervised similarity, but the final thesis-facing edge schema is still to be decided.

For controlled skills, the full-artifact condition uses the authored benchmark `SKILL.md`. For imported public-gold targets, the intended source artifact is the upstream publicly available `SKILL.md`; normalised import wrappers are retained for provenance and bookkeeping rather than as the public full-artifact source. The thesis does not assume that public skill authors always write clean fields. The representation layer is allowed to extract or normalise fields from prose. This is important because the public skill audit shows that many selection signals are present but not always explicitly labelled.

The current fixed RQ1 operational-information groups are task/use condition, input/precondition, output/artifact, workflow/procedure, dependency/resource/platform, boundary/not-for, and success/verification criteria where available. These are not all simple positive retrieval text. Task/use condition, input, output, and workflow are primary disambiguation fields, but task/use condition is partly an intent signal: it is highly discriminative when prompt and skill wording align, and less robust when the requested intent is paraphrased. Dependency and boundary are conditional feasibility, authority, scope, or guardrail fields. The dependency/resource suite makes this distinction explicit for capability compatibility: the

field is strong when the request or routing context contains a matching platform, tool, API, runtime, permission, version, or resource, but generic package-download prose has little routing value by itself. The boundary/not-for suite makes a parallel distinction: the field is strong when the request states an exclusion and weaker when the request only implies an authority boundary. Hierarchy, DAG, and graph relations are RQ2 strategy directions rather than settled RQ1 field groups unless a specific experiment maps them to operational information explicitly.

5.8 Retrieval and Reranking Methods

The method families are:

M0 progressive disclosure.: The main agent sees compact skill cards and decides which full skill documents to load. This is the normal-agent baseline, but it becomes expensive at large scale.

M1 flat lexical retrieval.: BM25 or TF-IDF over R1. This is a control for lexical cueing rather than a proposed final architecture.

M2 dense retrieval.: Embedding retrieval over R1, R2, or full artifacts. Current frozen-v0.4 provider experiments use Qwen `text-embedding-v4` and the released SkillRouter embedding model. MiniLM is retained mainly as a local construction and validation model.

M3 structured-card retrieval.: Retrieval over serialised R2 procedural fields. This tests whether procedural information helps when exposed in a structured card.

M4 tree routing.: A planned hierarchy-based architecture: route from root to family or cluster, then to skill. This tests scale efficiency and branch-exclusion failure. The branch information to expose is still TBD.

M5 graph retrieval.: A planned graph-based architecture over R4 edges. This tests whether explicit relations help beyond serialised text. The edge schema and the exact information carried by those edges are still TBD and must avoid benchmark-derived leakage.

M6-v0 diagnostic schema reranking.: The current deterministic local schema reranker. It uses weighted lexical overlap over extracted fields. It is transparent and useful for diagnosis, but too crude to be the final structure-aware method.

M6-v1 field-aware rerank.: The current first prototype of the proposed structure-aware method. It parses the request into fields and compares those fields against extracted skill fields.

M6-v2 semantic field-aware rerank.: A no-rewrite semantic field matcher. It keeps the first-stage candidate set fixed, embeds selector-visible skill fields separately, and compares the user request to those fields with an embedding model.

M7/M8 provider retrieval and reranking.: Provider or hosted-model embedding retrieval and neural reranking. Current frozen-v0.4 implementations include Qwen embeddings with Qwen reranking and SkillRouter embeddings with SkillRouter reranking. SkillRouter prompt-level artifacts have been recovered for detailed failure analysis.

5.9 M6-v0 Diagnostic Reranker

The existing schema reranker scores lexical overlap between the user request and skill fields. It assigns weight to skill name, description, use conditions, output shape, preconditions, workflow, dependencies/resources, and a negative boundary penalty.

This method is intentionally simple. It tests whether extracted procedural fields contain useful information, but it has limitations:

- it does not parse the user request into structured requirements;
- it uses lexical overlap rather than semantic field matching;
- it can overweight tool or resource words;
- it does not reliably detect broad parent/router skills;
- it treats extracted public-skill fields as clean even when extraction is noisy.

Therefore, M6-v0 should be reported as diagnostic evidence, not as the final proposed retrieval architecture.

5.10 Existing M6-v1 Field-Aware Diagnostic Prototype

The M6-v1 method better matches the thesis claim. Its pipeline is:

- (1) A first-stage retriever returns a top-50 or top-100 candidate set.
- (2) A request parser extracts request-side fields: intended action, input or precondition, desired output, workflow requirement, tool/platform requirement, explicit exclusion, and success criterion.

- (3) The reranker compares request fields to R2/R3 skill fields.
- (4) The final score combines first-stage retrieval score with field matches and penalties.

A transparent starting score is:

```
score =  
    0.30 * normalized_first_stage_score  
+ 0.20 * use_when_match  
+ 0.20 * output_match  
+ 0.15 * workflow_match  
+ 0.10 * input_precondition_match  
+ 0.05 * success_criterion_match  
+ conditional_dependency_bonus  
- boundary_mismatch_penalty  
- hierarchy_router_penalty
```

The purpose is not to hand-tune the benchmark until this score wins. The purpose is to test whether extracted selection information improves selection when it is used as field-to-field evidence rather than appended as plain text.

The implemented prototype should be interpreted as M6-v1-local, a lexical field-aware prototype. It is stronger than M6-v0 because it includes request-side field extraction and candidate-recall decomposition. However, it is not yet a robust semantic field matcher. The request parser uses cue-based extraction, and the field matcher mainly uses token overlap rather than learned semantic entailment or contradiction detection. A later version could replace the matcher with a stronger neural or LLM-assisted field matcher while keeping the same information taxonomy.

This distinction matters for interpretation. If M6-v1-local improves a controlled result, the evidence supports the value of the extracted fields and field-to-field matching under a transparent scoring rule. It does not yet prove that the scoring rule is the best possible structure-aware architecture. Conversely, if M6-v1-local fails on public-authored skills, that failure may come from noisy field extraction, weak request parsing, lexical mismatch, public-skill breadth, or duplicated public skills. These sources must be separated before making a general claim about structure-aware retrieval.

The current M6-v1-local analysis therefore records candidate recall separately from final top-1. A first-stage failure means the gold skill never entered the reranker candidate set. A reranking failure means the gold skill was present but not placed first. This separation is important because dense retrievers may be good high-recall candidate generators even when final ordering remains poor.

5.11 Existing M6-v2 Semantic Field-Aware Diagnostic Prototype

M6-v2 is the existing semantic field-matching prototype from the frozen-v0.4 exploratory pipeline. It keeps the same representation taxonomy as M6-v1 but replaces lexical field overlap with semantic field matching. It is not the new RQ2a field-aware selector because it is coupled to a first-stage score, selected field sets, calibrated weights, request parsing, and the older representations. Its reusable embedding, chunking, aggregation, and cache components will be adapted under the stricter RQ2a contract described above.

For each first-stage candidate, M6-v2 embeds the candidate's task/use conditions, input or precondition fields, output fields, workflow fields, dependency/resource fields, and boundary fields separately. The current implementation compares either the raw positive request against each field, or a field-specific request span against the corresponding skill field. Multi-item fields such as workflow are chunked and aggregated by maximum or top-two average similarity rather than by summing all matches, so longer skills are not automatically rewarded for containing more text.

The generic score form is:

```
score =
    alpha * normalized_first_stage_score
+ (1 - alpha) * normalized_semantic_field_score
+ explicit_exclusion_alignment_bonus
- boundary_or_dependency_mismatch_penalty
- broad_router_penalty
```

The parameter `alpha` controls whether M6-v2 behaves mainly as a first-stage-preserving reranker or as a field-dominant reranker. Field weights are tuned only as a calibration or ablation question. A full-set best weight should not be treated as a final result unless it is selected on a development split and

evaluated on held-out prompts. The thesis therefore reports M6-v2 both as an interpretability diagnostic and as a candidate method requiring further calibration.

For the main fixed M6-v2 condition, the same weights are used across all candidate sources, representations, and prompt strata. The fixed score uses a light first-stage prior and a stronger semantic-field term:

```
score =
    0.30 * normalized_first_stage_score
+ 0.70 * normalized_semantic_field_score
+ 0.25 * boundary_or_dependency_adjustment
```

The fixed semantic-field weights are:

- task/use condition: 0.50;
- output/artifact: 0.20;
- workflow/procedure: 0.15;
- input/precondition: 0.10;
- dependency/resource: 0.05.

This setting is intentionally not optimised separately per model or stratum. It tests whether the same explicit procedural-information policy transfers across candidate generators.

5.12 Historical Baselines and Optional Architecture Record

This section records the earlier broad method programme. These methods remain useful exploratory baselines, but the active RQ2a core is the matched-content BM25, Qwen single-vector, cross-encoder, and field-aware comparison described above.

SkillRouter-style models are necessary strong baselines because they represent the most directly related skill-specific retrieval family. However, they should be introduced as a baseline after the benchmark has already established the scaling and semantic-confusability problem. The fair comparison is not simply “SkillRouter versus our method.” The fair comparison asks whether a skill-specific dense retriever and reranker still benefits from, or is challenged by, the proposed field-aware information.

M4 tree routing and M5 graph retrieval are outside the current core. They introduce hierarchy or relation information not isolated by RQ1 and must not be implemented as part of RQ2a. They may be revisited only as separately motivated future work.

5.13 Historical SkillRouter Hybrid Experiment

To compare the proposed field-aware reranker against a strong skill-specific neural baseline, the thesis should also evaluate a SkillRouter hybrid pipeline:

- (1) SkillRouter’s released embedding model, `pipizhao/SkillRouter-Embedding-0.6B`, embeds full skill artifacts and the user request.
- (2) The embedding model produces a first-stage ranked candidate list.
- (3) Either SkillRouter’s released reranker, `pipizhao/SkillRouter-Reranker-0.6B`, reranks top-20 candidates, or the M6-v1-local field-aware reranker reranks top-20 or top-100 candidates.

The frozen-v0.4 local benchmark has already been run for SkillRouter embeddings and SkillRouter reranking over R1, R2, and full artifacts, and for SkillRouter-first-stage plus M6-v1 diagnostic reranking over matched candidate budgets. A remote Hugging Face Jobs container remains useful for reproducibility and larger reruns, but it is no longer a prerequisite for the existing frozen-v0.4 SkillRouter rows.

This design gives a comparison between two second-stage selection policies over the same strong first-stage retriever: learned SkillRouter reranking versus explicit procedural field matching. Candidate budget must be held fixed for a direct reranker comparison. Current frozen-v0.4 SkillRouter metrics, SkillRouter-to-M6-v1 diagnostic rows, and prompt-level artifacts are available for R1, R2, and full artifacts. The remaining work is interpretation, failure analysis, and cost/candidate-budget reporting. This local frozen-v0.4 track is distinct from the external SkillRouter-Eval-Core verification track, where full-tier I3C FTS/BM25 retrieval has now been run but fixed-candidate I3C reranking remains future work.

5.14 Execution Venue and Provider Boundary

The execution venue is recorded as part of each method configuration. This avoids conflating a local smoke test, a paid provider API run, a Codex-based extraction pass, and a hosted GPU retrieval matrix.

Experiment family	Execution venue	Boundary rule
Local validation, representation export, BM25/TF-IDF, SQLite FTS, aggregation, and thesis re-build	Local repository checkout	These are CPU/file operations and may be run locally. They must still report benchmark version, prompt stratum, information layer, and output artifact.
Qwen embedding and Qwen reranking	Local orchestration with Dash-Scope/Qwen API calls	Model computation and cost are provider-side. Runs require API credentials and should be reported as provider/API conditions, not local model inference.
DeepSeek/API I3M extraction	Local orchestration with paid provider API calls	This is a paid-model feasibility/pass attempt for model-parsed I3. It must not be relabelled as I3C.
ChatGPT/Codex I3C extraction	Codex subagents over assigned artifact text	Workers use the I3C extraction prompt, produce JSONL rows, and validate exact evidence spans. This is the practical thesis-facing extraction route, not a provider reranker run.
SkillRouter smoke test	Local subset only	Local SkillRouter execution is allowed only to verify code/model loading on tiny subsets. It is not a full result row.
SkillRouter full matrices and external SkillRouter-Eval-Core neural reruns	Hugging Face Jobs GPU or equivalent hosted GPU	The released SkillRouter embedder/reranker full-scale runs should not run on the laptop. Snapshot inputs and uploaded result artifacts are part of the method record.

TABLE 5.2: Execution venue boundaries for reproducible method records.

For the external SkillRouter-Eval-Core neural matrix, the current reproducible pattern is to package the relevant benchmark snapshot, upload it to the Hugging Face dataset repository, submit a Hugging Face Jobs GPU run, and persist the resulting Markdown/JSON artifacts back to the same dataset repository. Local commands may prepare or inspect the snapshot, but the full SkillRouter embedding and reranking computation is a hosted GPU experiment.

5.15 M4 Tree Routing

This section is retained as a historical architecture plan. M4 is not part of the active RQ2a protocol and has not been evaluated.

M4 represents skills in a hierarchy:

```
root -> family -> optional cluster -> skill
```

The method first routes the request to one or more families, then ranks only skills inside the chosen branch or branches. It should report:

- family top-1 accuracy;
- family top-n recall;
- final skill top-1/top-5/MRR;
- gold-excluded-by-branch rate;
- candidate reduction ratio;
- context and latency cost.

M4 mainly tests scalability and early-routing risk. It is not the central method for discovering which procedural fields matter.

5.16 M5 Graph Retrieval

This section is retained as a historical architecture plan. M5 is not part of the active RQ2a protocol and has not been evaluated.

M5 uses a heterogeneous graph over skills and information nodes. The current R4 export includes typed edges such as:

- belongs_to_family;
- triggered_by;
- avoid_when;
- has_precondition;
- has_workflow_step;
- produces_output;
- requires_dependency;
- has_resource_signal;
- uses_resource;
- derived_from_public_skill.

Two graph variants are planned. M5a is graph-only retrieval: the request is parsed into requirement nodes and matched to adjacent skill nodes. M5b is hybrid dense-plus-graph reranking: a dense retriever first returns a candidate set, and graph scoring reranks only that subgraph.

Not all edge types should contribute to ranking. For example, `derived_from_public_skill` is provenance evidence for audit and reporting, not retrieval evidence. Similarly, benchmark-derived relations such as manually listed alternatives should not be used by M5, because they would leak evaluation structure into the selector.

Graph edge ablations should test:

- family only;
- trigger only;
- trigger plus output;
- trigger plus output plus workflow;
- adding input/precondition;
- adding dependency/resource;
- adding avoid penalties;
- full graph.

This allows the thesis to distinguish an information claim from an architecture claim. If M5 improves over M6-v1, explicit relation structure may be useful. If M5 does not improve, the thesis can still argue that the fields matter while graph structure may be unnecessary overhead.

5.17 Evaluation Metrics

For current RQ2a, the primary metric is paired top-1 accuracy over fixed three-candidate sets, with the cluster as the inferential unit. MRR, rank margin, near-neighbour confusion, per-field and prompt-variant accuracy, selector-visible tokens, and latency are secondary. Candidate recall is not an RQ2a outcome because the gold skill is guaranteed to be present.

RQ2b uses the following full-library metrics. It reports the strict single-gold outcomes separately from older historical acceptable-alternative metrics:

- **Strict top-1 accuracy:** whether the top-ranked skill is the gold skill.
- **Strict top-k recall:** whether the gold skill appears within the top-k candidate set.
- **MRR@20:** how highly the strict gold skill is ranked within the retrieval cut-off.

- **Recall@20:** whether the B1 full-library ranking includes the strict gold in the candidate set available to B2.
- **Conditional reranked Hit@1:** strict top-1 after a Qwen or SkillRouter reranker has reordered the fixed B1 Top-20 list.

Strict scoring tests whether the selector recovers the intended frozen benchmark target. RQ2b deliberately does not convert historically discussed acceptable alternatives into a post-hoc second endpoint. Public-gold results are reported as their own stratum because imported skills can be broader, duplicated, or differently authored. A strict-label miss may therefore be a candidate miss, ordering error, broad parent/router selection, duplicate-like alternative, or an ambiguity that the present strict endpoint does not resolve; it is not direct evidence of downstream task failure.

The efficiency metrics are candidate budget, representation length, selector-visible token cost, latency, provider API calls, and cache behaviour.

Qualitative failure analysis records first-stage exclusion, reranker misweighting, boundary or negation failure, broad parent/router selection, public-skill duplication, underspecified request ambiguity, and acceptable-alternative ambiguity.

5.17.1 Historical External SkillRouter-Eval-Core Metric Alignment

The external SkillRouter-Eval-Core track uses related but not identical metrics. The local frozen-v0.4 benchmark usually has a single strict gold skill per prompt, plus optional acceptable alternatives. SkillRouter-Eval-Core tasks can require multiple core skills and include graded relevance labels. Therefore, the external track is reported with a compatible but multi-skill metric view:

- **Hit@1** is the closest analogue to strict top-1: the top-ranked skill must be one of the core gold skills.
- **MRR@10** is directly comparable as the reciprocal rank of the first core gold skill.
- **Recall@k** is not the same as local single-gold top-k accuracy when a task has multiple required skills; it measures the fraction of all core gold skills retrieved in the top-k list.
- **FullCoverage@k** is a stricter multi-skill analogue of top-k success: all core gold skills must appear in the top-k list.

- **nDCG@10** uses SkillRouter’s graded relevance annotations and has no exact local strict-gold equivalent.
- **Acceptable-alternative scoring** is not currently available for SkillRouter-Eval-Core unless an additional acceptable-alternative layer is manually defined.

The external runner is therefore configured to report the thesis cutoffs where useful, especially Recall@5 and FullCoverage@5, while keeping the original SkillRouter multi-skill metrics. External results should be used as portability evidence, not merged directly into the local controlled/public-gold tables.

5.17.2 Historical External Corpus and Prompt Audit

Because SkillRouter-Eval-Core is used as an external portability check, its corpus and prompts are also audited for structural comparability with the local benchmark. This audit is not a retrieval result. It measures whether skill bodies contain explicit markdown structure and whether prompts provide direct routing cues.

The audit shows that SkillRouter-Eval-Core is not uniformly an I3-style structured corpus: only about 27% of all skills, and about 27% of the scored gold-name subset, satisfy the strict rich-schema test used in the audit. However, the corpus is highly documentation-like. In the full SkillRouter corpus, 95.0% of skill bodies have an H1 heading and 78.8% contain procedural signal. In the scored gold-name subset, 98.5% have an H1 heading and 79.2% contain procedural signal. Frequent section headings include `overview`, `best practices`, `when to use`, `when to use this skill`, `workflow`, `quick start`, `dependencies`, and `resources`. This means that full-body retrieval on the external benchmark often searches clean markdown documentation sections rather than arbitrary unstructured text.

The prompt distribution is also different. The 75 scored SkillRouter tasks have a mean length of 198.4 rough whitespace tokens and a median length of 169. They contain file or path references in 97.3% of tasks, explicit input cues in 92.0%, explicit output cues in 85.3%, and numbered steps in 40.0%. By contrast, the local controlled prompts average 36.8 rough tokens with a median of 25, and are designed with near-neighbour alternatives. The external benchmark is therefore useful for portability and scale, but it should not be treated as the same difficulty regime as the local controlled semantic-confusability benchmark.

5.18 Cost and Trade-off Accounting

RQ2a records representation-construction time, document-scoring preparation time, cache size, selector-visible length, cold and warm query latency, and provider calls. RQ2b extends this ledger to B1 document/index time, B1 query time, selector-visible corpus tokens, and B2 fixed-Top-20 reranking work. The two studies have different candidate regimes, so their absolute latency figures are not pooled.

One-time representation construction cost.: This includes parsing skill files, extracting I3 fields, validating evidence spans, and building serialised representation files. For I3C this can be expensive, because it may involve Codex/ChatGPT-style parsing and QA. This cost is paid when the skill library changes, not on every user query.

One-time indexing or embedding cost.: Lexical indexes and embedding vectors can be pre-computed. Full documents are expensive here because every full artifact must be indexed or embedded; I3 fields may reduce this cost by reducing the number of selector-visible tokens. If embeddings are cached, this is mostly an amortised library-maintenance cost.

Per-query selection cost.: This includes query embedding or lexical lookup, first-stage retrieval latency, reranking calls, candidate-budget size, and any main-agent context used for final skill loading. This cost is paid for every user request.

For a library with N skills and Q future user queries, the rough total cost of a representation r can be written as:

```
total_cost(r, Q) =
  extraction_cost(r)
+ index_or_embedding_cost(r)
+ Q * per_query_selection_cost(r)
+ Q * expected_full_doc_hydration_cost(r)
```

This formula states the intended amortisation model, but RQ2b reports only costs that were durably recorded. It measures representation-size, B1 document/index, B1 query, and B2 reranking quantities. The V3 extraction record contains 2,433 source rows, 57 source chunks, and 1,455,632 proxy input tokens, but it lacks a durable comparable extraction wall-time or provider invoice. The thesis therefore does not calculate a total I3C construction cost or a break-even query count. It can state only that I3C

reduces the observed B1 selector surface and document/index burden relative to I2, while its strict-label accuracy may be lower in a given stratum and retriever.

5.19 RQ2a Statistical Comparison Plan

All RQ2a comparisons are paired because every representation and selector sees the same query, three candidate identities, and gold label. Prompt variants from one cluster are not treated as independent observations. The primary contrast is `same-facts-fielded` minus `same-facts-flat` under Qwen single-vector selection. The effect is reported in top-1 percentage points with a 95% cluster-bootstrap interval based on at least 10,000 resamples.

The required replications apply the same fielded-versus-flat contrast under BM25 and the fixed cross-encoder. Key secondary contrasts are fielded versus prose, fielded versus diluted-2x, fielded versus shared-only, and field-aware Qwen versus single-vector Qwen on identical fielded text. Holm correction is applied to the predeclared secondary family. MRR, margin, field strata, and prompt-variant strata are reported as supporting analyses. Mixed-effects logistic regression with representation as a fixed effect and cluster as a random intercept is a sensitivity analysis, not the primary estimator.

Technical completion and scientific support are reported separately. A run passes technically only if every expected row exists, candidate identities align, decisive text is not truncated, caches and model versions are recorded, and no confirmatory score was used to retune a method. Subject to supervisor approval, the organisation or field-aware hypotheses receive practical support when their paired top-1 gain is at least three percentage points and the corresponding cluster-bootstrap interval excludes zero. Null or negative outcomes are retained when the controls pass.

5.20 Unreviewed Downstream Validation Option

Downstream validation is neither part of active RQ2a nor an approved RQ2b requirement. If separately reviewed later, a smaller sample of prompts could be selected after offline retrieval behaviour is stable. The main agent would receive selected candidate skills under fixed conditions, generate task artifacts, and be graded for task success, artifact correctness, evidence use, and misuse of wrong skills.

This step is necessary before claiming that retrieval improvements directly improve final agent task performance. Until then, the thesis should frame current results as retrieval and representation evidence rather than full end-to-end agent reliability evidence.

5.21 Reproducibility

Reproducibility is supported by storing prompts, skill artifacts, representation exports, selector scripts, cached provider outputs, validation reports, public-skill audit outputs, and manual adjudication notes in the project repository. Provider-based experiments should report model names, candidate budgets, cache status, and approximate cost or latency where available.

CHAPTER 6

Results

6.1 Current Result Scope

The current result set answers two research questions through four completed experiments, alongside a set of historical exploratory records. RQ1 uses two complementary interventions: seven AI-assisted, researcher-directed and researcher-reviewed controlled near-neighbour suites, each with 50 clusters and field-specific prompt variants; and a source-grounded public-document intervention that removes cited information from exact original documents. RQ2a is a matched-content fixed-candidate mechanism study. RQ2b is a source-grounded, strict-gold full-library study over 2,433 skills and 381 prompts (243 controlled and 138 public-gold). Historical representation and retrieval matrices based on frozen benchmark version `benchmark-v0.4-2026-06-16` are retained separately for provenance and portability context.

Section 6.2 reports the completed RQ1 controlled and public-artifact evidence. Section 6.3 reports the completed matched-content RQ2a confirmatory study over 280 untouched clusters, 600 prompts, 22 conditions, and 13,200 aligned decisions. Section 6.4 reports the completed full-library RQ2b validation, including first-stage candidate retrieval, fixed-Top-20 reranking, failure decomposition, and recorded cost. Later frozen-v0.4, M6, and external SkillRouter sections are retained as exploratory, diagnostic, feasibility, or portability evidence; they must not be read as the final RQ2a or RQ2b answer. Graph/tree retrieval and downstream execution remain outside the active scope.

6.2 RQ1 Results: Controlled and Public-Artifact Evidence

6.2.1 Public-Skill Primary-Document Prevalence

Before testing whether a field changes routing, we checked whether the proposed information types actually recur in public skill artifacts. A frozen source frame of 29,292 canonical public primary skill documents was reviewed in 31 batches by a Codex gpt-5.6-terra agent using `xhigh` reasoning. The agent used a simple binary rule: a field received 1 when it was expressed anywhere in the complete primary document and 0 otherwise. The review included YAML frontmatter, including `name` and `description`, and every Markdown section, table, and example; it did not inspect attached scripts, references, or other auxiliary files. Thus the census measures information present in the agent-visible primary skill document, not only information stated in the Markdown body.

Candidate information field	Documents present	Share of 29,292
Use condition	23,634	80.68%
Input / precondition	20,618	70.39%
Output / artifact	23,363	79.76%
Workflow / procedure	24,412	83.34%
Success / verification	20,968	71.58%
Boundary / not-for	19,543	66.72%
Dependency / resource	22,064	75.32%
Few-shot example	19,013	64.91%

TABLE 6.1: Model-assisted binary review of the complete primary document for 29,292 frozen public skills. Metadata and Markdown body are both in scope; auxiliary files are not.

The census supplies descriptive support for field selection: the categories motivated by the literature also recur in independently authored public skill documents, at different but substantial frequencies. It does not show that every field occurs in every skill, that a field originates specifically in metadata or body text, or that field presence improves retrieval. Those usefulness claims remain for the controlled field-isolation and public original-document removal experiments below. The review is model-assisted rather than independent human annotation and was completed in 31 recorded batches, so it is used as descriptive context rather than as a public annotation dataset or quality certification.

6.2.2 Controlled Field-Isolation Results

The clearest current RQ1 results are the completed field-isolation suites for use condition, input/precondition, output/artifact, dependency/resource, boundary/not-for, success/verification, and workflow/procedure. These results are reported before the broader retrieval matrix because they answer the thesis-first question directly: whether one operational field helps separate semantically similar skills when non-target information is controlled.

Each suite contains 50 near-neighbour clusters. Most suites use one direct and one paraphrased prompt per cluster; dependency/resource replaces the paraphrase with a contextual variant, and boundary/not-for adds an implicit-authority variant. Within each cluster, the three sibling skills share all non-target fields. The hidden condition exposes only their shared context, so they are intentionally tied. The exposed condition presents the same context plus one candidate-specific field value. Top-1 and MRR use tie-aware scoring. BM25 and Qwen `text-embedding-v4` are the primary retrievers; DeepSeek and Codex selectors are secondary reasoning checks. The suites were created with AI assistance under a researcher-defined rubric and mechanically checked. The researcher then reviewed and approved all 350 rows with the labels visible; this was not an independent blinded review.

For both retrievers, Top-1 improves after the target field is shown in all seven suites. The suites are separate authored test sets, however, and are not used to rank fields against one another. Input/precondition remains robust under paraphrase, reaching 91.0% BM25 and 94.0% Qwen Top-1. Output/artifact shows a clear retriever difference: its paraphrased rows reach 53.3% with BM25 and 92.0% with Qwen. Use condition is also less robust under paraphrase than under direct trigger wording. Dependency/resource is highly effective when it names a concrete capability such as a required tool, platform, credential, runtime, or version; this claim does not extend to generic installation commands.

The more conditional fields remain informative but harder to match. Boundary/not-for falls to 62.0% BM25 and 58.0% Qwen under implicit-authority prompts. Paraphrased success/verification rows reach 48.0% for both retrievers, and workflow/procedure reaches 56.0% BM25 and 50.0% Qwen under paraphrase. These values show sensitivity to how the request expresses scope, acceptance gates, and procedure order. Full prompt-subset Top-1 and MRR tables are retained in the experiment artifacts; Table 6.2 provides the main controlled summary.

The target-field values were not forced to equal length. In particular, gold boundary and dependency values are often longer or more detailed than their alternatives. This does not undo the paired result that

exposing candidate-specific information changes routing, but the controlled effect should be read as the value of the field content as written, not as a length-normalised estimate of field meaning alone.

6.2.3 Controlled Result Summary and Uncertainty

Prompt variants are paired within a cluster. The analysis therefore averages variants within each of the 50 clusters and uses 10,000 cluster-bootstrap resamples of the change in Top-1 accuracy between the field-hidden and field-shown conditions. Table 6.2 shows a positive 95% cluster interval for every field/retriever row. These intervals describe variation across each frozen 50-cluster suite; they do not rank the seven fields or establish effects for arbitrary public skills.

Field	BM25 with field	BM25 improvement, 95% CI	Qwen with field	Qwen improvement, 95% CI
Use condition	.845	+.512 [+ .447, +.572]	.910	+.577 [+ .517, +.627]
Input/precondition	.955	+.622 [+ .582, +.657]	.970	+.637 [+ .597, +.667]
Output/artifact	.767	+.433 [+ .373, +.493]	.960	+.627 [+ .587, +.657]
Dependency/resource	1.000	+.667 [+ .667, +.667]	.990	+.657 [+ .637, +.667]
Boundary/not-for	.833	+.500 [+ .440, +.553]	.747	+.413 [+ .327, +.493]
Success/verification	.740	+.407 [+ .337, +.477]	.590	+.257 [+ .137, +.377]
Workflow/procedure	.780	+.447 [+ .377, +.517]	.620	+.287 [+ .187, +.377]

TABLE 6.2: Controlled-RQ1 Top-1 accuracy after the target field is shown, and improvement over the tie-aware field-hidden baseline of .333. Intervals are 95% cluster-bootstrap intervals for the improvement. Each field has its own 50-cluster suite, so rows should be read within field rather than as a formal cross-field ranking.

We also measured the score gap between the gold skill and the highest-scoring alternative. This gap increased in all fourteen field–retriever rows, and every 95% cluster-bootstrap interval was above zero. In other words, showing the field usually moved the gold skill farther ahead of its strongest competitor; the Top-1 result was not produced only by a deterministic tie break. This does not identify an objectively closest alternative: both alternatives were authored as plausible siblings, but the units do not claim that one is semantically closer than the other.

The reasoning-selector checks support the same limited conclusion: once the field is visible, DeepSeek selects the gold skill in 88–100% of cases across the seven suites and Codex in 96.7–100%. These are supporting checks of whether the distinctions are understandable, not the primary retrieval result.

An additional negative-control suite tests `examples/tests`. It is not counted as an eighth operational field. In that suite, all operational fields are shared by sibling skills and only the `examples/tests` section differs. Generic prompts have no true operational gold because all siblings can perform the task;

they test whether examples create arbitrary attraction. Rare exact-match prompts deliberately resemble one stored example and estimate the upper bound of example attraction. The corrected multi-method result supports treating examples/tests as workflow-support context rather than a primary routing signal. On generic prompts, exposing examples does not improve retrieval: BM25 moves from 33.3% hidden to 22.7%, TF-IDF remains approximately tied at 33.0%, and Qwen embedding moves from 33.3% to 28.0%. DeepSeek and Codex subagent selectors both mark generic rows as ambiguous when examples are visible. On rare exact-match prompts, all methods reach 100.0% selection of the example-matching candidate. This shows that examples can strongly steer routing when the request copies or closely resembles a stored example, but that effect is better described as example-attraction or possible misguidance than as proof of a distinct skill capability.

6.2.4 Public Original-Document Removal

The second RQ1 experiment applies the same question to preserved public skill documents. A source-tracked pool of 1,099 documents supplies 82 candidate groups. Before scoring, the sources, near-neighbour groups, cited field lines, prompts, and gold skills were checked. For each group and field, the selector sees either the intact documents or the same documents after every cited line carrying that field has been removed from all candidates. The implementation blanks those line contents and leaves the remaining text unchanged. The candidates, prompt, source hashes, and gold skill remain fixed. An edited skill may be incomplete or non-executable because this experiment tests routing after information removal, not execution after editing. The researcher approved all 194 scored gold-label cases and all 402 scored removals with the labels and document edits visible; this was not an independent blinded review. Appendix B.7 gives concrete examples of both RQ1 interventions; those examples were selected without using their retrieval outcomes.

The final scoring set contains 1,078 field-specific cases, 2,156 direct/paraphrased requests, and 4,312 ranking rows per retriever. Each field has 49–69 eligible candidate groups. Requests that belong to the same group are averaged first, and 10,000 paired bootstrap samples estimate the uncertainty in the difference between the intact and field-removed documents. Qwen uses cached `text-embedding-v4` document/query vectors and BM25 uses local lexical scores; their score margins use different scales and are therefore reported separately.

Field	Source compositions	BM25 Full-minus-removed Hit@1, 95% CI	Qwen Full-minus-removed Hit@1, 95% CI
Use condition	57	+.114 [+.067, +.167]	+.058 [+.031, +.092]
Input/precondition	53	+.098 [+.051, +.149]	+.016 [−.017, +.052]
Output/artifact	49	+.122 [+.065, +.187]	+.003 [−.036, +.036]
Workflow/procedure	63	+.151 [+.086, +.215]	+.040 [−.024, +.100]
Success/verification	59	+.117 [+.059, +.182]	+.027 [+.005, +.054]
Boundary/not-for	69	+.024 [−.014, +.064]	−.022 [−.054, +.008]
Dependency/resource	52	+.091 [+.046, +.139]	+.013 [−.048, +.067]

TABLE 6.3: Public original-document single-field removal. Positive Full-minus-removed values mean that removing the cited field lowers strict Hit@1. Each field has a separate clean-only source-composition denominator.

Use condition and success/verification have a stable positive strict Hit@1 effect in both retrievers. Input/precondition, output/artifact, workflow/procedure, and dependency/resource have stable positive effects under BM25 but not under Qwen; they are therefore retriever-conditional in this public-document population. Boundary/not-for has no stable positive strict Top-1 effect in either retriever. These results do not say that a field is the sole human rationale for choosing a skill. Natural documents can express one operational fact across titles, descriptions, procedures, examples, and resource requirements. They show the more limited result that removing the cited field lines can weaken agreement with the frozen strict label.

We also removed three predefined groups of fields by deleting all cited lines belonging to any field in the group. These supporting tests show the effect of removing the combined information; they cannot assign that effect to one member field or show that individual effects add together.

Removed group	Families / compositions	BM25 Full-minus-removed Hit@1, 95% CI	Qwen Full-minus-removed Hit@1, 95% CI
Task specification: use + input + output	99 / 37	+.266 [+.196, +.336]	+.088 [+.005, +.162]
Execution/verification: workflow + success	138 / 51	+.176 [+.094, +.262]	+.074 [+.007, +.141]
Applicability/capability: boundary + dependency	132 / 50	+.136 [+.087, +.186]	+.014 [−.043, +.062]

TABLE 6.4: Public original-document joint-group removal. The groups delete the exact union of their member field lines, so their effects cannot be assigned to a single field.

Task specification and execution/verification show a stable loss after removal under both retrievers. Applicability/capability does so only under BM25. The two experiments therefore answer different parts of RQ1. The controlled experiment shows that each tested field can distinguish near-neighbour

skills when it is the only candidate difference. The public experiment shows which fields and field groups still matter when the same information is scattered through natural documents. The earlier field-card experiment is kept only as a historical sensitivity check and is not combined with these results. RQ2 next asks how different representations and retrievers use the available information.

6.3 RQ2a Confirmatory Matched-Content Results

RQ2a is complete through the reviewed-fact build, 70-cluster development split, frozen field-aware aggregation choice, untouched 280-cluster confirmatory split, paired statistical analysis, and cost ledger. The confirmatory split contains 600 prompt variants. Every condition uses the same query, the same three sibling candidates, and the same reviewed operational propositions; only representation form or scoring mechanism changes. The full matrix contains 22 conditions and 13,200 unique aligned condition–prompt decisions.

6.3.1 Full Condition Matrix

Table 6.5 reports cluster-weighted tie-adjusted top-1, MRR, strict near-neighbour confusion, and top-tie rate. The `shared-only` negative control intentionally gives all three siblings byte-identical text. Its top-1 of $1/3$ and MRR of $(1 + 1/2 + 1/3)/3 = 0.611$ therefore verify unbiased tie handling rather than model failure.

Table 6.5: Complete RQ2a confirmatory condition matrix. All rows contain the same 600 prompts and fixed three-candidate decisions unless the selector has fewer preregistered diagnostic representations.

Selector	Representation	Top-1	MRR	Confusion	Tie rate
BM25	diluted 1x	0.864	0.920	0.127	0.023
BM25	diluted 2x	0.863	0.919	0.128	0.023
BM25	diluted 4x	0.864	0.920	0.127	0.023
BM25	fielded	0.864	0.921	0.127	0.023
BM25	flat	0.843	0.911	0.147	0.023
BM25	order-controlled	0.864	0.921	0.127	0.023
BM25	prose	0.840	0.908	0.152	0.023
BM25	shared-only	0.333	0.611	0.000	1.000
Qwen field-aware, uniform top two	fielded	0.823	0.901	0.145	0.057
Qwen single-vector	diluted 1x	0.778	0.876	0.228	0.000

Selector	Representation	Top-1	MRR	Confusion	Tie rate
Qwen single-vector	diluted 2x	0.778	0.872	0.232	0.000
Qwen single-vector	diluted 4x	0.763	0.867	0.247	0.000
Qwen single-vector	fielded	0.757	0.863	0.252	0.000
Qwen single-vector	flat	0.824	0.902	0.185	0.000
Qwen single-vector	order-controlled	0.790	0.880	0.223	0.000
Qwen single-vector	prose	0.782	0.873	0.233	0.000
Qwen single-vector	shared-only	0.333	0.611	0.000	1.000
SkillRouter cross-encoder	diluted 2x	0.942	0.967	0.063	0.000
SkillRouter cross-encoder	fielded	0.951	0.973	0.053	0.000
SkillRouter cross-encoder	flat	0.955	0.975	0.048	0.000
SkillRouter cross-encoder	prose	0.946	0.969	0.058	0.000
SkillRouter cross-encoder	shared-only	0.333	0.611	0.000	1.000

The content positive controls are large for all three selector families. Relative to `shared-only`, fielded top-1 rises by 42.4 percentage points for Qwen, 53.1 points for BM25, and 61.8 points for SkillRouter. Candidate-specific operational facts are therefore necessary for resolving these controlled near-neighbour decisions.

6.3.2 Preregistered Paired Contrasts

Table 6.6 reports the frozen top-1 contrasts. Prompt-level differences are paired within cluster, and inference uses 10,000 cluster bootstraps plus 10,000 paired cluster sign flips. The eight secondary rows receive Holm correction. Practical support requires a gain of at least three percentage points and a 95% interval excluding zero.

Contrast	Δ Top-1	95% cluster CI	Holm p	Practical
Primary: Qwen fielded — flat	−0.067	[−0.099, −0.036]	—	No
BM25 fielded — flat	+0.021	[+0.010, +0.033]	0.0040	No
SkillRouter fielded — flat	−0.004	[−0.014, +0.006]	0.5240	No
Qwen fielded — prose	−0.025	[−0.060, +0.010]	0.3188	No
Field-aware — Qwen fielded	+0.066	[+0.021, +0.111]	0.0204	Yes
Qwen fielded — diluted 2x	−0.021	[−0.036, −0.006]	0.0303	No
Qwen fielded — shared-only	+0.424	[+0.380, +0.467]	0.0008	Yes
BM25 fielded — shared-only	+0.531	[+0.506, +0.555]	0.0008	Yes
SkillRouter fielded — shared-only	+0.618	[+0.599, +0.635]	0.0008	Yes

TABLE 6.6: Preregistered RQ2a top-1 contrasts. The primary contrast has raw paired sign-flip $p = 0.0002$ and was not included in the secondary Holm family. “Practical” applies the frozen positive three-percentage-point threshold.

The primary positive organisation hypothesis is rejected. With exactly the same reviewed propositions, Qwen single-vector selection performs 6.7 points worse on fielded text than on flat text. BM25 gains 2.1 points from the headings, but the effect is below the frozen practical threshold. SkillRouter is effectively insensitive to fielded versus flat form in this matrix. Explicit field headings are therefore not an independent or universal source of routing accuracy.

The frozen Qwen field-aware rule gains 6.6 points over Qwen fielded single-vector selection and satisfies both inferential and practical criteria. Its top-1 of 0.823 is nevertheless almost identical to Qwen flat at 0.824. Field-wise scoring recovers the penalty created when fielded text is pooled into one vector; it does not establish superiority over the strongest matched Qwen serialisation.

6.3.3 Field Heterogeneity

The aggregate recovery is not uniform across fields. Table 6.7 reports descriptive slices; they were not separately multiplicity-tested and are not seven additional confirmatory claims.

Field	Qwen flat	Qwen fielded	Field-aware	FA — fielded
Use condition	0.925	0.888	0.938	+0.050
Input/precondition	0.913	0.888	0.900	+0.013
Output/artifact	0.963	0.950	0.917	−0.033
Workflow/procedure	0.638	0.600	0.658	+0.058
Dependency/resource	0.963	0.963	0.771	−0.192
Boundary/not-for	0.683	0.625	0.739	+0.114
Success/verification	0.688	0.388	0.838	+0.450

TABLE 6.7: Descriptive RQ2a field-level top-1 slices for Qwen. Uniform-top-two field scoring especially recovers success/verification and boundary cases, but harms dependency/resource and output/artifact cases.

This heterogeneity explains why one global field aggregation rule is not universally optimal. Uniform top two helps where the decisive evidence is conditional or distributed, especially success/verification, boundary, workflow, and use-condition cases. It can suppress a single already-strong signal, as in dependency/resource and output/artifact. Field-adaptive policies are therefore a future hypothesis rather than a conclusion from the frozen confirmatory design.

6.3.4 Cost and Latency

Primary run	API calls	Provider tokens	Construction	Online/prompt	Total wall
BM25	0	0	0.000 s	0.001100 s	0.945 s
Qwen single-vector	590	1,175,784	1,085.293 s	0.002869 s	1,087.359 s
Qwen field-aware	77	13,583	203.259 s	0.002398 s	204.760 s
SkillRouter cross-encoder	0	0	0.000 s	11.099921 s	6,666.917 s

TABLE 6.8: RQ2a primary construction and online costs. Qwen construction is amortisable because exact candidate vectors persist. SkillRouter reports novel pair inference, not a warm score-cache lookup.

All 600 query embeddings were exact cache hits, so confirmatory execution sent no query text. The two Qwen conditions made 667 successful calls for 6,661 new candidate or field-component texts and 1,189,367 provider tokens. The dated standard-list-price estimate is USD 0.0833. Warm verification made zero provider calls and zero new SkillRouter forward passes.

The accuracy–cost result is therefore architectural. Qwen independently encodes documents once and performs millisecond cosine comparisons at query time. SkillRouter jointly re-encodes every novel query–candidate pair and reaches the strongest fixed-candidate accuracy, 0.955 on flat text, but requires about 11.10 seconds per prompt in this environment. The result does not establish a universal winner: it quantifies the trade-off within a fixed three-candidate setting.

6.4 RQ2b Full-Library Retrieval And Reranking Results

RQ2b evaluates the frozen V3 library of 2,433 source-grounded skills against 381 strict-gold prompts. It uses 243 controlled prompts in 25 semantic clusters and 138 public-gold prompts in 61 semantic clusters. Each result is strict agreement with the frozen intended gold label. The full-library metric therefore measures retrieval behaviour under this benchmark endpoint, not unique task correctness or downstream task success.

6.4.1 First-Stage Retrieval

Table 6.9 gives the complete B1 cluster-macro Hit@1 matrix. I1 uses the native name and description, I2 uses the original full body, I3-flat uses retained I3 evidence without headings, and I3C uses the same retained evidence with operational-field headings. The three retrievers see exactly the same source library and prompts for each representation.

Stratum	B1 retriever	I1	I2	I3-flat	I3C
Controlled	BM25	0.559	0.535	0.452	0.453
Controlled	Qwen	0.351	0.367	0.358	0.361
Controlled	SkillRouter	0.700	0.656	0.679	0.682
Public-gold	BM25	0.597	0.644	0.512	0.504
Public-gold	Qwen	0.612	0.686	0.560	0.538
Public-gold	SkillRouter	0.761	0.806	0.752	0.746

TABLE 6.9: RQ2b V3 B1 strict Hit@1, aggregated equally by semantic cluster. Qwen is maximum cosine over document chunks; SkillRouter is its native bi-encoder first stage.

The main result is not a universal I3C accuracy gain. On the public-gold stratum, I2 exceeds I3C under every first-stage retriever: by 14.0 percentage points for BM25, 14.8 points for Qwen, and 5.9 points for SkillRouter. The corresponding post-hoc paired cluster-bootstrap intervals for I3C minus I2 are $[-0.247, -0.033]$, $[-0.224, -0.080]$, and $[-0.140, +0.021]$. These intervals describe the fixed benchmark; no p-value or significance threshold is used as a thesis claim gate. On controlled prompts the pattern is mixed: I1 is strongest for BM25 and SkillRouter, while I2 is narrowly strongest for Qwen.

Candidate recall shows why later reranking cannot be treated as a general repair. Table 6.10 compares I2 and I3C Recall@20 before any reranker acts. I2 has higher Recall@20 in all six reported B1 rows. A reranker receives only the already-retrieved Top-20 candidates, so it cannot recover a strict gold that is absent from that list.

Stratum	B1 retriever	I2 Recall@20	I3C Recall@20	I3C – I2
Controlled	BM25	0.929	0.880	-0.049
Controlled	Qwen	0.773	0.715	-0.058
Controlled	SkillRouter	0.980	0.974	-0.006
Public-gold	BM25	0.984	0.865	-0.119
Public-gold	Qwen	1.000	0.921	-0.079
Public-gold	SkillRouter	0.992	0.975	-0.017

TABLE 6.10: RQ2b B1 strict Recall@20 for I2 and I3C. Recall@20 is candidate inclusion before B2, not a final reranked score.

6.4.2 Fixed-Top-20 Reranking

For each per-retriever matrix of 1,524 B1 conditions (4,572 conditions across all three B1 retrievers), B2 reorders the same persisted Top-20 candidate identities with Qwen `qwen3-rerank` or the SkillRouter cross-encoder. Each reranker therefore produces 4,572 rows, or 9,144 B2 rows in total. Table 6.11 reports all 24 reranked cluster-macro Hit@1 cells. The table is conditional on B1 candidate generation: it

compares final order under a fixed candidate set rather than an end-to-end method that can independently change recall.

Stratum	B1 retriever	B2 reranker	I1	I2	I3-flat	I3C
Controlled	BM25	Qwen	0.741	0.685	0.712	0.713
Controlled	Qwen	Qwen	0.588	0.553	0.577	0.592
Controlled	SkillRouter	Qwen	0.785	0.723	0.765	0.777
Controlled	BM25	SkillRouter	0.689	0.682	0.673	0.689
Controlled	Qwen	SkillRouter	0.561	0.570	0.538	0.560
Controlled	SkillRouter	SkillRouter	0.717	0.714	0.725	0.737
Public-gold	BM25	Qwen	0.771	0.812	0.718	0.718
Public-gold	Qwen	Qwen	0.812	0.842	0.734	0.758
Public-gold	SkillRouter	Qwen	0.845	0.834	0.785	0.785
Public-gold	BM25	SkillRouter	0.734	0.779	0.715	0.712
Public-gold	Qwen	SkillRouter	0.772	0.796	0.747	0.752
Public-gold	SkillRouter	SkillRouter	0.772	0.771	0.790	0.785

TABLE 6.11: RQ2b V3 B2 strict Hit@1 after fixed-Top-20 reranking. Each reranker reorders the corresponding B1 candidate list only.

Reranking changes order but does not remove the representation trade-off. On public-gold, I3C remains below I2 in five of the six retriever–reranker combinations. The one exception is SkillRouter reranking after SkillRouter B1, where I3C is 0.785 and I2 is 0.771. The descriptive failure decomposition is consistent with the Recall@20 result: I3C-minus-I2 candidate-miss rates are non-negative in every comparison. This is compatible with I2 retaining candidate-inclusion detail that I3C omits, but does not identify a particular omitted span as the cause of an individual miss.

6.4.3 Representation Cost And Availability

Table 6.12 reports the measured B1 representation and document/index burden for I2 and I3C. I3C consistently reduces the selector-visible corpus by 72–76% and recorded document/index time by 64–79%. The observed gains are meaningful for a maintained library, but they do not include the full one-time extraction cost: the V3 archive records source rows, chunks, and proxy tokens but lacks a durable comparable extraction wall-time or provider invoice. No total I3C cost or break-even query volume is claimed.

B1 retriever	I2 tokens	I3C tokens	Token reduction	I2 doc/index s	I3C doc/index s
BM25	825269	228222	72.3%	0.233	0.083
Qwen	1455632	349639	76.0%	966.577	324.292
SkillRouter	1458065	352072	75.9%	80.286	16.967

TABLE 6.12: Measured RQ2b B1 cost frontier for I2 and I3C. Tokens are selector-visible corpus tokens; document/index time is one-time work for the measured run.

B4 tests a possible explanation for I1’s competitive results without changing any retrieval row. It classifies a retained I3 item as description-visible only when its original source text, after whitespace folding, appears in the frozen I1 metadata description. Only 75 of 19,394 retained items (0.4%) are description-visible across the full corpus. Among the 381 strict-gold prompt records, only 4 of 755 retained items (0.5%) are description-visible; controlled records have 0 of 639 and public-gold records have 4 of 116. Thus, the result does not support the simple explanation that I1 succeeds because it already duplicates the retained I3 operational content. It remains possible that I1 carries coarse task/category cues or that fixed benchmark and extraction/formatting properties affect matching, but B4 does not causally distinguish these alternatives.

The bounded RQ2b answer is therefore an accuracy–compression trade-off. I3C reduces the first-stage selector surface and document/index burden relative to I2, but does not improve strict-label accuracy universally under the tested flat first-stage retrievers or fixed-list rerankers. RQ1 establishes that the operational fields can be discriminative in controlled near-neighbour cases; RQ2b shows that extracting and serialising those fields alone is not sufficient to guarantee a better full-library selector.

6.5 Historical Frozen-v0.4 Benchmark Validation

Table 6.13 summarises the frozen-v0.4 benchmark validation status.

These validation results support the use of the benchmark as a semantic-confusability stress test. The controlled stratum is stronger for causal comparisons because the gold skill and alternatives were deliberately designed around procedural distinctions. The public-gold stratum is messier but important for external validity because its target skills are publicly available skill artifacts rather than only authored controlled skills.

6.6 Historical Frozen-v0.4 Method Coverage

Table 6.14 records which methods have been run against frozen v0.4. The blank entries are deliberate methodological gaps, not negative results.

For public-gold full-artifact comparisons, final reporting should use the upstream public `SKILL.md` artifacts. Wrapper-based public import rows are treated as implementation-history rows until rerun against the original public artifacts.

Check	Frozen-v0.4 result
Composition	245 controlled prompts, 144 public-gold prompts, 12 low-information prompts, 2433 skills.
Controlled integrity	PASS: 245 prompts, 0 missing references.
Controlled procedural distinctness	PASS: 245/245 prompts; 830/830 gold/alternative pairs.
Controlled prompt-specific alignment	238/245 prompts pass; strict top-1 among listed candidates 236/245; acceptable top-1 241/245.
Controlled semantic confusability	PASS: 243/245 prompts. Two residual observability prompts remain documented.
Controlled prompt leakage	PASS: 0 critical exact-name leaks and 0 high-risk leaks.
Public-gold integrity	PASS: 144 prompts, 0 missing references.
Public-gold procedural distinctness	PASS: 144/144 prompts; 575/575 gold/alternative pairs.
Public-gold prompt-specific alignment	131/144 prompts pass; strict top-1 among listed candidates 118/144; acceptable top-1 138/144.
Public-gold semantic confusability	PASS: 144/144 prompts.
Public-gold prompt leakage	PASS: 0 critical exact-name leaks and 0 high-risk leaks.

TABLE 6.13: Frozen-v0.4 benchmark validation status.

Family	Method	Controlled	Public-gold
Local lexical	BM25 and TF-IDF over R1	Done	Done
Local embedding	MiniLM description, full skill, and schema	Done	Done
Structured lexical	TF-IDF/BM25 over schema text	Done	Done
Schema rerank	M6-v0 local schema rerank	Done	Done
Qwen embedding	R1, R2, and full skill	Done	Done
Qwen rerank	Qwen top-20 rerank over R1, R2, and full skill	Done	Done
Local schema over Qwen	Qwen first stage plus local-schema top-20	Done	Done
M6-v1 local	BM25, TF-IDF, TF-IDF schema, and Qwen-full first stages; top-20/top-100	Done	Done
Field ablations	28 local ablation conditions	Done	Done
SkillRouter/Hugging Face	SkillRouter embedding and reranker over R1/R2/full	Done	Done
M6-v1 + SkillRouter	SkillRouter top-20 first stage plus field-aware rerank	Done	Done
M4 tree routing	Hierarchical route then rank	TBD	TBD
M5 graph retrieval	Graph filter/rerank over R4 edges	TBD	TBD
M0 full-scale progressive disclosure	Main-agent baseline at 2433-skill scale	TBD	TBD
Downstream validation	Agent task success after retrieval	TBD	TBD

TABLE 6.14: Frozen-v0.4 method coverage. Historical SkillRouter runs are excluded; current SkillRouter summary metrics are from the frozen-v0.4 rerun.

6.7 Historical Exploratory Representation-First Matrix

The result tables below follow the intended comparison standard: first group by the *representation layer*, then list the retrieval or reranking method, candidate budget, and metrics. This prevents an invalid comparison where a method appears better simply because it was given a richer or noisier skill representation.

6.7.1 Controlled Stratum

Table 6.15 reports the 245-prompt controlled stratum. The controlled stratum is the cleanest evidence for representation effects because the gold skills and alternatives were designed around explicit procedural distinctions.

Representation	Method	Budget	Top-1	Top-5	MRR
R1 flat metadata	M1 BM25 flat	all	55.9%	87.4%	0.693
R1 flat metadata	M1 TF-IDF flat	all	53.9%	85.3%	0.677
R1 flat metadata	M2a MiniLM description	all	38.8%	77.1%	0.550
R1 flat metadata	Qwen embedding only	all	34.3%	62.5%	0.470
R1 flat metadata	Qwen embedding + Qwen rerank	20	51.4%	74.7%	0.612
R1 flat metadata	Qwen embedding + local schema rerank	20	44.9%	74.7%	0.585
R1 flat metadata	SkillRouter embedding only	all	62.9%	93.5%	0.763
R1 flat metadata	SkillRouter embedding + SkillRouter rerank	20	65.7%	94.7%	0.780
R2/R3 structured procedural	M3 TF-IDF schema	all	64.5%	93.1%	0.772
R2/R3 structured procedural	M3 MiniLM schema	all	49.8%	83.7%	0.650
R2/R3 structured procedural	Qwen embedding only, R2	all	40.0%	71.8%	0.538
R2/R3 structured procedural	Qwen embedding + Qwen rerank, R2	20	58.0%	80.4%	0.679
R2/R3 structured procedural	Qwen embedding + local schema rerank, R2	20	52.6%	79.6%	0.647
R2/R3 structured procedural	SkillRouter embedding only, R2	all	64.5%	94.7%	0.783
R2/R3 structured procedural	SkillRouter embedding + SkillRouter rerank, R2	20	71.0%	96.7%	0.823
Full SKILL, mcd artifact	M2b MiniLM full skill	all	41.2%	76.7%	0.567
Full SKILL, mcd artifact	Qwen embedding only	all	44.1%	75.5%	0.575
Full SKILL, mcd artifact	Qwen embedding + Qwen rerank	20	58.8%	80.8%	0.686
Full SKILL, mcd artifact	Qwen embedding + local schema rerank	20	53.9%	80.8%	0.661
Full SKILL, mcd artifact	SkillRouter embedding only	all	59.2%	91.4%	0.737
Full SKILL, mcd artifact	SkillRouter embedding + SkillRouter rerank	20	71.8%	95.9%	0.827

Table 6.15: Controlled frozen-v0.4 representation-layer results. Field-aware hybrid rerankers are reported separately because they vary both representation and architecture.

The controlled results show a consistent representation-layer signal. Under Qwen embedding-only, R2 improves over R1 from 34.3% to 40.0% top-1, while full skill text reaches 44.1%. Under Qwen reranking, R2 improves over R1 from 51.4% to 58.0%, and full text is only slightly higher at 58.8%. SkillRouter is much stronger than Qwen under the same R1/R2/full comparison. Its R2 reranked result reaches 71.0% top-1, while full skill text plus SkillRouter reranking reaches 71.8%. This suggests that structured procedural representation can recover much of the useful signal in the full skill artifact while exposing less irrelevant text. The table does not yet prove that any specific structure-aware architecture is best; it isolates what happens when different amounts and kinds of skill information are exposed to comparable retrieval methods.

6.7.2 Public-Gold Stratum

Table 6.16 reports the 144-prompt public-gold stratum. Public-gold results are reported with strict-gold top-1 and gold-or-acceptable top-1 because public skills contain duplicates, broad parent artifacts, and near-equivalent alternatives.

Representation	Method	Budget	Strict top-1	Accept top-1	Strict top-5	MRR
R1 flat metadata	M1 BM25 flat	all	55.6%	65.3%	81.9%	0.673
R1 flat metadata	M1 TF-IDF flat	all	56.9%	64.6%	86.8%	0.697
R1 flat metadata	M2a MiniLM description	all	56.9%	63.2%	79.9%	0.670
R1 flat metadata	Qwen embedding only	all	57.6%	66.0%	82.6%	0.687
R1 flat metadata	Qwen embedding + Qwen rerank	20	74.3%	81.9%	92.4%	0.820
R1 flat metadata	Qwen embedding + local schema rerank	20	50.0%	61.1%	83.3%	0.640
R1 flat metadata	SkillRouter embedding only	all	68.8%	77.1%	93.8%	0.794
R1 flat metadata	SkillRouter embedding + SkillRouter rerank	20	70.8%	79.2%	92.4%	0.804
R2/R3 structured procedural	M3 TF-IDF schema	all	48.6%	55.6%	81.2%	0.629
R2/R3 structured procedural	M3 MiniLM schema	all	48.6%	58.3%	79.9%	0.621
R2/R3 structured procedural	Qwen embedding only, R2	all	63.2%	72.2%	86.8%	0.738
R2/R3 structured procedural	Qwen embedding + Qwen rerank, R2	20	72.9%	82.6%	97.9%	0.832
R2/R3 structured procedural	Qwen embedding + local schema rerank, R2	20	49.3%	61.8%	88.9%	0.657
R2/R3 structured procedural	SkillRouter embedding only, R2	all	75.0%	81.2%	94.4%	0.831

Representation	Method	Budget	Strict top-1	Accept top-1	Strict top-5	MRR
R2/R3 structured procedural	SkillRouter embedding + SkillRouter rerank, R2	20	71.5%	78.5%	93.1%	0.805
Full SKILL.md artifact	M2b MiniLM full skill	all	40.3%	48.6%	61.1%	0.504
Full SKILL.md artifact	Qwen embedding only	all	36.1%	47.9%	59.7%	0.480
Full SKILL.md artifact	Qwen embedding + Qwen rerank	20	65.3%	71.5%	79.9%	0.716
Full SKILL.md artifact	Qwen embedding + local schema rerank	20	33.3%	46.5%	68.8%	0.483
Full SKILL.md artifact	SkillRouter embedding only	all	75.0%	82.6%	93.1%	0.833
Full SKILL.md artifact	SkillRouter embedding + SkillRouter rerank	20	70.1%	76.4%	92.4%	0.804

Table 6.16: Public-gold frozen-v0.4 representation-layer results. Field-aware hybrid rerankers are reported separately because they vary both representation and architecture.

The public-gold results are more mixed. R2 is best for Qwen embedding-only retrieval, improving strict top-1 over R1 from 57.6% to 63.2%. However, R1 plus Qwen reranking has the highest Qwen strict top-1 at 74.3%, while R2 plus Qwen reranking has the highest Qwen MRR and strict top-5. SkillRouter embedding is strongest on public-gold before reranking: both R2 and full artifact representations reach 75.0% strict top-1. However, SkillRouter reranking reduces strict top-1 on R2 and full text, which suggests that the neural reranker is not uniformly helpful on messy public-authored skills. Full skill text is a negative result for Qwen but not for SkillRouter, so the public-gold stratum shows an interaction between representation and model family rather than a single global ranking of representations.

6.7.3 Crossed Lexical Representation Control

To check that the representation effect is not only an artifact of dense embedding models, BM25 and TF-IDF were also run over the same three representation layers: R1 flat metadata, R2 structured procedural cards, and full SKILL.md text. These are not the strongest expected final methods, but they provide a clean control because the retriever is held fixed while the representation changes.

Stratum	Retriever	R1 top-1	R2 top-1	Full top-1	R1 cand. R@20	R2 cand. R@20
Controlled	BM25	55.9%	62.9%	63.3%	94.3%	98.0%
Controlled	TF-IDF	53.9%	64.5%	61.2%	91.8%	97.1%
Public-gold	BM25	55.6%	53.5%	34.0%	87.5%	97.9%
Public-gold	TF-IDF	56.9%	48.6%	36.8%	92.4%	95.1%

TABLE 6.17: Crossed lexical representation control on frozen v0.4. Public-gold values are strict-gold top-1. Candidate recall is measured before any second-stage reranking.

The controlled lexical control gives direct evidence that R2 contains useful selection information: with the retriever fixed, R2 improves BM25 top-1 by 7.0 percentage points and TF-IDF top-1 by 10.6 percentage points over R1. Public-gold behaves differently. R2 improves candidate recall, especially for BM25, but does not improve strict top-1 under simple lexical ordering. This indicates that structured procedural fields can help surface the correct public skill into the shortlist, while final ordering still depends on a stronger reranker that can handle provider cues, duplicate capabilities, broad public artifacts, and noisy public prose.

6.7.4 Reranking and Field-Aware Diagnostics

The previous tables isolate representation-layer effects. Table 6.18 moves to architecture-level diagnostics. These rows should not be read as pure representation-layer comparisons because they change the candidate generator, reranker, selected field set, and candidate budget. Their purpose is to test whether explicit procedural fields can help once a first-stage retriever has already produced a candidate subset.

Stratum	First stage	Reranker / field use	Bgt.	Top-1	Top-5	MRR
Controlled	BM25 flat	M6-v0 schema rerank	20	60.8%	90.2%	0.743
Controlled	TF-IDF flat	M6-v0 schema rerank	20	63.3%	88.6%	0.749
Controlled	MiniLM full skill	M6-v0 schema rerank	20	54.7%	83.3%	0.676
Controlled	BM25 flat	M6-v1 core-boundary field rerank	20	67.8%	92.7%	0.792
Controlled	BM25 flat	M6-v1 core-boundary field rerank	100	68.6%	94.7%	0.805
Controlled	TF-IDF flat	M6-v1 core-boundary field rerank	20	69.8%	91.4%	0.793
Controlled	TF-IDF flat	M6-v1 core field rerank	100	69.8%	94.3%	0.809
Controlled	TF-IDF schema	M6-v1 core-boundary field rerank	20	69.8%	97.1%	0.818
Controlled	TF-IDF schema	M6-v1 core-boundary field rerank	100	69.0%	96.3%	0.814
Controlled	Qwen full skill	M6-v1 core field rerank	20	57.6%	81.2%	0.684
Controlled	Qwen full skill	M6-v1 core-boundary field rerank	100	64.5%	91.4%	0.764
Public-gold	BM25 flat	M6-v0 schema rerank	20	43.1%	80.6%	0.585
Public-gold	TF-IDF flat	M6-v0 schema rerank	20	45.8%	84.7%	0.612
Public-gold	MiniLM full skill	M6-v0 schema rerank	20	40.3%	69.4%	0.523
Public-gold	BM25 flat	M6-v1 task-field rerank	20	63.9%	84.0%	0.724
Public-gold	BM25 flat	M6-v1 task-field rerank	100	63.9%	84.0%	0.724
Public-gold	TF-IDF flat	M6-v1 task-field rerank	20	68.1%	87.5%	0.763
Public-gold	TF-IDF flat	M6-v1 task-field rerank	100	67.4%	86.8%	0.756
Public-gold	TF-IDF schema	M6-v1 task-field rerank	20	66.7%	87.5%	0.756
Public-gold	TF-IDF schema	M6-v1 task-field rerank	100	65.3%	87.5%	0.743
Public-gold	Qwen full skill	M6-v1 task-field rerank	20	56.2%	73.6%	0.641
Public-gold	Qwen full skill	M6-v1 task-field rerank	100	59.0%	81.9%	0.695

Stratum	First stage	Reranker / field use	Bgt.	Top-1	Top-5	MRR
Table 6.18: Frozen-v0.4 architecture-level field-aware diagnostics. Public-gold top-1 and top-5 are strict-gold metrics.						

The diagnostic results show that field-aware reranking is promising but not yet a solved method. On controlled prompts, M6-v1 improves substantially over M6-v0, and the strongest current local setting is TF-IDF schema first-stage retrieval with M6-v1 core-boundary reranking at 69.8% top-1 and 97.1% top-5. On public-gold prompts, the best M6-v1 setting reaches 68.1% strict top-1, below Qwen R1+Qwen reranking at 74.3%, but with useful evidence that explicit task fields can recover part of the procedural signal. This supports treating M6-v1 as a diagnostic method for field usefulness rather than the final proposed architecture.

6.7.5 Candidate Recall Companion Table

Reranked systems have two possible failure points: the first-stage retriever may fail to include the gold skill, or the reranker may include it but order it incorrectly. Candidate recall separates these failure modes. Candidate R@20 measures whether the gold skill was present in the top-20 candidate set before reranking. Candidate R@100 gives the same diagnostic for wider candidate budgets. Conditional top-1 is the final top-1 accuracy among prompts where the gold skill was available to the reranker.

Stratum	Method	Bgt.	Cand. R@20	Cand. R@100	Final top-1	Cond. top-1	Main bottleneck
Controlled	Qwen R1 embedding + Qwen rerank	20	75.5%	—	51.4%	—	Recall + order
Controlled	Qwen R2 embedding + Qwen rerank	20	80.8%	—	58.0%	—	Recall + order
Controlled	Qwen full embedding + Qwen rerank	20	82.4%	—	58.8%	—	Recall + order
Controlled	TF-IDF schema + M6-v1 core-boundary rerank	20	97.1%	98.4%	69.8%	71.9%	Ordering
Controlled	Qwen full + M6-v1 core-boundary rerank	100	82.5%	93.5%	64.5%	69.0%	First-stage + order
Public-gold	Qwen R1 embedding + Qwen rerank	20	93.8%	—	74.3%	—	Ordering
Public-gold	Qwen R2 embedding + Qwen rerank	20	99.3%	—	72.9%	—	Ordering
Public-gold	Qwen full embedding + Qwen rerank	20	81.2%	—	65.3%	—	Candidate recall
Public-gold	TF-IDF flat + M6-v1 task-field rerank	20	92.4%	97.9%	68.1%	73.7%	Ordering
Public-gold	Qwen full + M6-v1 task-field rerank	100	81.2%	95.8%	59.0%	61.6%	First-stage + order

TABLE 6.19: Candidate-recall companion table for selected reranked methods on frozen v0.4. Candidate recall is measured before reranking.

The candidate-recall table changes the interpretation of several results. For controlled TF-IDF schema plus M6-v1, the gold skill is almost always present in the candidate set, so remaining errors are mostly reranking errors. For Qwen full-skill first stages, candidate recall is lower, especially on public-gold,

meaning the full artifact representation can fail before the reranker has a chance to correct the ranking. This is why final accuracy alone is insufficient: it hides whether the system needs a better first-stage representation, a better reranker, or both.

6.7.6 Global Field-Ablation Diagnostics

The frozen-v0.4 global field ablations are retained as diagnostic support, not as the primary RQ1 causal evidence. Unlike the controlled RQ1 suites, these rows do not hold all non-target fields constant around one labelled target field. They show how different field subsets behave under local lexical scoring across the broader controlled and public-gold strata. Table 6.20 shows the strongest current ablation conditions for each stratum.

Stratum	Ablation	Top-1	Top-5	MRR
Controlled	Output artifacts, BM25	66.5%	95.1%	0.788
Controlled	Input/preconditions, BM25	66.1%	93.5%	0.782
Controlled	Workflow/procedure, TF-IDF	65.7%	92.2%	0.777
Controlled	Use conditions, BM25	64.9%	93.9%	0.777
Public-gold	Description only, TF-IDF	56.2%	88.2%	0.695
Public-gold	Description only, BM25	55.6%	86.8%	0.690
Public-gold	Dependencies/resources, BM25	53.5%	87.5%	0.676
Public-gold	Use conditions, TF-IDF	52.8%	83.3%	0.663

TABLE 6.20: Selected global field-ablation diagnostics on frozen v0.4. These rows support interpretation but should not replace the controlled RQ1 field-isolation suites for individual-field causality.

The global diagnostics support different claims for the two strata. Controlled prompts benefit from procedural fields such as outputs, preconditions, workflow, and use conditions. Public-gold prompts remain strongly driven by descriptions. This is consistent with the RQ1 conclusion, but less clean: field usefulness depends on whether the decisive signal is exposed, whether it is extracted cleanly, and whether the matcher can compare it robustly. Therefore these diagnostics should be reported as RQ2-adjacent support for representation design, while individual field claims should come from the controlled RQ1 hidden-versus-exposed suites and the later public original-document removal study.

6.8 Historical External SkillRouter-Eval-Core Verification

The external SkillRouter-Eval-Core track tests whether the information-layer claim transfers beyond the local frozen-v0.4 benchmark. It is not a replacement for the controlled and public-gold strata above. The external benchmark has 75 scored non-generic tasks, an Easy pool of 78,361 public skills, and a Hard pool of 79,141 skills. Many tasks require multiple skills, so the main metrics are Hit@1, MRR@10, Recall@k, FullCoverage@k, and nDCG@10 rather than only single-gold top-1.

Table 6.21 reports the primary full-tier SQLite FTS5 BM25 information-layer ablation after the cleaned I3C V2 full-all extraction. The I3C condition serialises only evidence-grounded selector fields from the cleaned I3C V2 artifact. It should therefore be compared against I1 metadata and I2 full-body artifacts under the same retriever.

Tier	Layer	Hit@1	MRR@10	nDCG@10	R@5	R@20	Full@5	Full@20	Tokens
Easy	I1 metadata	36.0%	0.448	0.328	39.1%	48.3%	26.7%	32.0%	5.49M
Easy	I2 full body	48.0%	0.558	0.412	48.5%	57.4%	34.7%	42.7%	152.67M
Easy	I3C V2 fields	53.3%	0.586	0.429	48.4%	60.5%	29.3%	44.0%	15.68M
Hard	I1 metadata	28.0%	0.373	0.278	34.4%	46.1%	24.0%	30.7%	5.54M
Hard	I2 full body	44.0%	0.525	0.392	47.3%	55.7%	33.3%	41.3%	153.16M
Hard	I3C V2 fields	45.3%	0.527	0.378	44.5%	59.2%	26.7%	40.0%	15.83M

TABLE 6.21: External SkillRouter-Eval-Core full-tier information-layer matrix using SQLite FTS5 BM25. R@k is multi-skill Recall@k; Full@k is whether all core gold skills are retrieved in the top-k.

On the Easy tier, I3C V2 is best on Hit@1, MRR@10, nDCG@10, Recall@20, and FullCoverage@20. On the Hard tier, I3C V2 narrowly beats I2 full body on Hit@1, MRR@10, and Recall@20, while I2 remains slightly stronger on FullCoverage@20. The cost trade-off is also substantial: I3C V2 uses about 15.8M approximate selector-visible tokens on the Hard tier, compared with 153.2M for I2 full body.

The result therefore supports the RQ1 claim that task, input, output, workflow, dependency, boundary, and success information can help when it is recovered cleanly. It does not show that local frozen-v0.4 I3C has been run. The local model-parsed artifact is I3M, produced through a paid DeepSeek/API route, and should remain labelled as a feasibility attempt rather than as the practical I3C condition.

6.8.1 External Corpus and Prompt Audit

The external SkillRouter-Eval-Core result should be interpreted with a corpus-difference caveat. A static audit of the SkillRouter skills and tasks shows that the external benchmark is more documentation-like

and more prompt-explicit than the local controlled benchmark. Table 6.22 summarises the key structural differences.

Corpus or prompt set	Rows	Median tokens	H1	Rich schema	Procedural
SkillRouter all skills	79141	726	95.0%	27.6%	78.8%
SkillRouter scored gold-name skills	533	885	98.5%	27.4%	79.2%
Local controlled gold+alternatives	240	260.5	100.0%	72.9%	73.3%

TABLE 6.22: Static skill-body audit for the external SkillRouter corpus and local controlled comparison set. Rich schema means at least four core selection categories are explicit headings, or at least three plus three numbered steps.

The most frequent SkillRouter H2 headings are also template-like. Across the full corpus the leading headings are overview (13,568), best practices (10,567), when to use (10,041), when to use this skill (9,642), examples (6,862), workflow (6,442), and quick start (6,382). In the scored gold-name subset, common headings include overview, best practices, quick start, dependencies, when to use this skill, and workflow. Thus, even though most SkillRouter rows do not satisfy the strict rich-schema definition, many full bodies are already clean markdown documentation with explicit routing sections.

Prompt set	Rows	Mean tok.	Median tok.	≥ 200 tok.	File/path	Input cue	Output cue
SkillRouter scored tasks	75	198.4	169	41.3%	97.3%	92.0%	85.3%
Local controlled prompts	245	36.8	25	0.4%	28.2%	18.4%	26.9%
Local public-gold prompts	144	30.1	29	0.0%	18.1%	21.5%	47.2%

TABLE 6.23: Prompt-side explicitness audit. Token counts are rough whitespace-token counts, not provider tokenizer counts.

This audit explains why the external result should be used as portability evidence rather than as a direct replacement for the local controlled benchmark. SkillRouter scored prompts are paragraph-sized task specifications with frequent file, input, output, and step cues. Local controlled prompts are shorter and intentionally include near-neighbour alternatives. Therefore, differences between local and external results may reflect both information-layer quality and benchmark difficulty.

6.9 Historical Frozen-v0.4 Method Results

6.9.1 SkillRouter and Hugging Face Results

SkillRouter frozen-v0.4 metrics and prompt-level artifacts are now available for R1, R2, and full-artifact representations. The detailed failure analysis is kept as a working note rather than inserted into this chapter.

Stratum	Representation	Method	Top-1	Top-5	MRR
Controlled	R1	Embedding only	62.9%	93.5%	0.763
Controlled	R1	SkillRouter rerank top-20	65.7%	94.7%	0.780
Controlled	R2	Embedding only	64.5%	94.7%	0.783
Controlled	R2	SkillRouter rerank top-20	71.0%	96.7%	0.823
Controlled	Full	Embedding only	59.2%	91.4%	0.737
Controlled	Full	SkillRouter rerank top-20	71.8%	95.9%	0.827
Public-gold	R1	Embedding only	68.8%	93.8%	0.794
Public-gold	R1	SkillRouter rerank top-20	70.8%	92.4%	0.804
Public-gold	R2	Embedding only	75.0%	94.4%	0.831
Public-gold	R2	SkillRouter rerank top-20	71.5%	93.1%	0.805
Public-gold	Full	Embedding only	75.0%	93.1%	0.833
Public-gold	Full	SkillRouter rerank top-20	70.1%	92.4%	0.804

TABLE 6.24: Frozen-v0.4 SkillRouter summary metrics from the hosted rerun. Prompt-level artifacts have been recovered for separate failure analysis.

The controlled SkillRouter result strengthens the main benchmark claim: even a skill-specific retriever and reranker does not solve the benchmark completely. R2 improves over R1 under both embedding-only and reranked settings, and full skill text helps most after reranking. On public-gold, SkillRouter embedding is strong, but reranking reduces strict top-1 for R2 and full text. This makes SkillRouter a strong baseline rather than a replacement for the representation question.

6.9.2 SkillRouter First Stage With M6-v1 Field-Aware Reranking

Table 6.25 compares two second-stage policies over the same SkillRouter top-20 candidate sets: the released SkillRouter neural reranker and the local M6-v1 lexical field-aware reranker. This is a fairer architecture comparison than comparing different first-stage retrievers, because the candidate generator, representation, prompt split, library size, and top-20 budget are held fixed.

The SkillRouter–M6-v1 comparison is useful but mixed. On controlled prompts, M6-v1 improves over embedding-only in all three representation conditions, showing that explicit field matching can use information in the candidate set. However, the released SkillRouter reranker remains slightly stronger than

Stratum	Rep.	Emb. only	SR rerank	M6-v1	M6 field set	Cand. R@20
Controlled	R1	62.9%	65.7%	67.8%	task/out./workflow	95.9%
Controlled	R2	64.5%	71.0%	70.2%	core + boundary	97.1%
Controlled	Full	59.2%	71.8%	68.6%	core + boundary	96.3%
Public-gold	R1	68.8%	70.8%	70.1%	task	96.5%
Public-gold	R2	75.0%	71.5%	72.2%	task	97.9%
Public-gold	Full	75.0%	70.1%	73.6%	task	97.9%

TABLE 6.25: SkillRouter top-20 first-stage candidates reranked by SkillRouter versus M6-v1 on frozen v0.4. Values are strict top-1 accuracy. M6-v1 uses the best field set within each row and is therefore a diagnostic upper view of the current lexical field-aware prototype.

M6-v1 for R2 and full representations. On public-gold prompts, the strongest SkillRouter embedding-only rows are already high, and both rerankers can reduce strict top-1. This suggests that public-authored skills often contain provider, platform, and task cues that make second-stage reranking brittle. The result supports the thesis claim that field-aware information is useful, but it also confirms that the current lexical M6-v1 matcher should be treated as diagnostic rather than final.

6.9.3 M6-v2 Semantic Field-Matching Calibration

M6-v2 is a no-rewrite semantic field-aware reranker. It embeds the raw user request and compares it against selector-visible skill fields. The first frozen-v0.4 implementation used a conservative score blend of 0.70 first-stage score and 0.30 semantic-field score. A later calibration sweep varied this blend and the penalty scale using existing candidate-level outputs. The sweep should be read as a diagnostic calibration result, not as a final tuned method, because full-set best weights can overfit the benchmark.

The thesis-facing fixed version now uses one global setting across all models and strata: 0.30 first-stage score, 0.70 semantic-field score, and a 0.25 penalty scale. The selected fixed internal field weights are task/use condition 0.50, output/artifact 0.20, workflow/procedure 0.15, input/precondition 0.10, and dependency/resource 0.05. Table 6.26 reports this fixed policy on the R2 structured representation. This is the clean method result; the later tuned tables are diagnostics.

As a boundary check, the task-heavy field-only version with zero first-stage score performs worse: Qwen controlled R2 34.3%, Qwen public-gold R2 47.2%, SkillRouter controlled R2 39.6%, and SkillRouter public-gold R2 50.7% strict top-1. This suggests that the explicit fields alone are not sufficient under the current no-rewrite matcher; a small first-stage prior is still useful for stabilising candidate order.

Candidate source	Stratum	Strict top-1	Accept top-1	Strict top-5	MRR
Qwen	Controlled R2	37.5%	38.0%	72.7%	0.524
Qwen	Public-gold R2	59.0%	67.4%	86.8%	0.709
SkillRouter	Controlled R2	54.3%	55.1%	91.4%	0.701
SkillRouter	Public-gold R2	68.1%	76.4%	91.7%	0.775

TABLE 6.26: M6-v2-fixed task-heavy field-weight result on R2. The same weights, blend, activation rule, and penalty scale are used in every row.

The task-heavy internal weighting was selected after a fixed-method ablation in which the outer scoring policy remained constant and only the internal field weights changed. Table 6.27 reports the average across the four main R2 conditions: Qwen controlled, Qwen public-gold, SkillRouter controlled, and SkillRouter public-gold.

Internal scheme	Avg top-1	Avg accept top-1	Avg top-5	Avg MRR	Worst top-1
Task-heavy	54.7%	59.2%	85.6%	0.677	37.5%
Dependency-balanced	53.4%	57.3%	83.7%	0.662	36.3%
Global core	51.9%	56.4%	83.0%	0.650	37.1%
Workflow-heavy	51.7%	56.6%	83.1%	0.650	37.1%
Balanced-input	50.8%	55.2%	82.8%	0.642	36.7%

TABLE 6.27: M6-v2 fixed internal-weight ablation on R2. The outer method is fixed at 0.30 first-stage score, 0.70 field score, and 0.25 penalty scale.

Candidate source	Stratum	Rep.	Dev-selected field set	Base wt.	Field wt.	Dev top-1	Test top-1
Qwen	Controlled	R2	task	0.55	0.45	41.7%	42.4%
Qwen	Public-gold	R2	task	0.30	0.70	72.7%	66.7%
SkillRouter	Controlled	R2	task	0.75	0.25	63.8%	64.4%
SkillRouter	Public-gold	R2	task	0.25	0.75	83.3%	66.7%

TABLE 6.28: M6-v2 fine blend calibration for selected R2 rows. Weights are selected on a deterministic development split and reported on the held-out split.

The calibration results suggest that the original M6-v2 first-stage weight was too conservative for some Qwen candidate sets, especially public-gold R2, where a field-heavy blend improves held-out top-1 relative to the initial M6-v2 task result. However, this pattern does not hold uniformly. SkillRouter controlled results remain more first-stage-heavy, which means explicit field scoring should act as calibrated evidence rather than a universal override. The next methodological step is therefore to tune M6-v2’s explicit field use: compare request-side input, output, workflow, and dependency evidence directly against the corresponding skill fields, instead of relying only on whole-query similarity to every field.

That field-specific M6-v2 pass has now been run as an ablation. Table 6.29 summarises the R2 rows because R2 is the main structured-representation condition. The fixed field-specific score compares request-side fields against matching skill-side fields, while the tuned rows recompute the semantic field score from saved per-field component similarities. The tuning sweep is narrow and should be treated as diagnostic.

Source	Stratum	Best tuned setting	Fixed top-1	Full-set top-1	Held-out top-1
Qwen	Controlled R2	core-boundary, base 0.90	40.0%	41.6%	40.7%
Qwen	Public-gold R2	task-only, base 0.25	61.8%	68.8%	66.7%
SkillRouter	Controlled R2	core-boundary, base 0.70	64.5%	66.1%	63.6%
SkillRouter	Public-gold R2	task-only, base 0.40	73.6%	76.4%	71.8%

TABLE 6.29: M6-v2 field-specific tuning on R2 after the focused explicit field-weight sweep. Fixed field-specific top-1 uses the best untuned field-specific field set; tuned full-set top-1 may overfit and is reported only as a calibration diagnostic.

The field-specific ablation clarifies the current limitation. Extra fields can improve full-set controlled R2 tuning in some rows, especially when boundary or dependency-aware field sets are used with SkillRouter candidates. However, held-out results remain weaker and the public-gold rows still mostly prefer task-only settings. This suggests that the proposed information fields are useful, but the current no-rewrite M6-v2 scorer is not yet sophisticated enough to decide reliably when output, workflow, dependency, or boundary fields should override the task-level match. A stronger future variant would need better request-field activation, contradiction handling, and validation-selected field weights.

6.9.4 M4 Tree Routing and M5 Graph Retrieval

Current frozen-v0.4 result: TBD. R4 graph edges have been exported, but tree routing and graph retrieval have not yet been evaluated. These should remain blank until implemented.

6.9.5 M0 Progressive Disclosure

Current frozen-v0.4 result: TBD. Existing M0 traces belong to older core conditions. Full-scale progressive disclosure has not been rerun against the 2433-skill frozen library.

6.9.6 Downstream Task Success

Current frozen-v0.4 result: TBD. No downstream task-success experiment has been completed for frozen v0.4. Current claims are therefore about retrieval quality, not final agent task success.

6.10 Historical Exploratory Interpretation

The frozen-v0.4 results support six exploratory observations that motivate, but do not answer, the new RQ2a:

- (1) **Representation matters under controlled confusability.** With Qwen held fixed, R2 structured representation improves over R1 flat representation on controlled prompts, and nearly matches full skill text under Qwen reranking.
- (2) **Skill-specific retrieval raises the ceiling but preserves the representation question.** SkillRouter substantially improves over generic Qwen on controlled prompts, but R2 and full artifact representations still behave differently under the same SkillRouter architecture.
- (3) **Full skill text is not automatically best.** On public-gold, full-skill Qwen embedding performs much worse than R1 or R2. More raw text can dilute the retrieval signal.
- (4) **Lexical candidate generation is a serious baseline.** M6-v1 with TF-IDF/schema or TF-IDF/flat first stages performs strongly on frozen v0.4. The thesis should not frame the comparison as only embedding versus structure.
- (5) **Public-gold is a harder external-validity check.** Public-gold results show that descriptions and provider/task cues remain powerful, while schema extraction and lexical field matching are brittle for messy public skills.
- (6) **The current field-aware method is diagnostic, not final.** M6-v1-local supports the value of field-aware matching, but a stronger semantic or LLM-assisted field matcher is needed before claiming a robust structure-aware retrieval method.

Overall, the current evidence supports an information-centred thesis, but the strongest claim should be phrased carefully: preserving operational selection information helps under controlled semantic confusion, but the way that information is extracted, represented, and matched determines whether the benefit transfers to public-authored skills.

Discussion

7.1 What the Current Results Suggest

RQ1 begins with a descriptive prevalence premise and then uses two complementary interventions to study operational-information routing. In a frozen pool of 29,292 public primary skill documents, each complete `SKILL.original.md` file—including its name, description, and Markdown content—received a model-assisted binary presence review. All eight candidate information types recur frequently: from 64.91% for few-shot examples to 83.34% for workflow/procedure; use condition, output/artifact, dependency/resource, and success/verification occur in 80.68%, 79.76%, 75.32%, and 71.58% of documents respectively. This establishes that the field candidates are not arbitrary additions to the thesis, but it is neither a claim about every public skill worldwide nor evidence that presence alone improves retrieval.

The controlled experiment keeps non-target information shared among three closely related skills and then shows one operational field. For both BM25 and Qwen, Top-1 improves after the field is shown in every field’s own suite. Robustness nevertheless depends on the prompt and retriever: paraphrased outputs are much harder for BM25, while implicit boundaries and paraphrased success or workflow requirements are difficult for both retrievers. Because the seven suites contain different authored cases, these results show that each field can help but do not establish a formal strongest-to-weakest ranking. The field values also vary naturally in length and detail, so the estimate concerns the information as written rather than field meaning after length normalisation. Examples/tests provide the negative control: generic examples do not create a stable distinction, whereas rare near-exact matches can attract a selector without proving a distinct capability.

The public original-document removal experiment tests whether the same information remains consequential when it is embedded in natural artifacts. It compares intact public source documents with the

same documents after every cited line for one field, or a defined field group, has been removed from all candidates. Use condition and success/verification have stable strict Hit@1 losses under both BM25 and Qwen. The task-specification and execution/verification groups also have stable losses under both retrievers. Input/precondition, output/artifact, workflow/procedure, dependency/resource, and applicability/capability are positive only under BM25; boundary/not-for has no stable positive result. Thus public documents carry both useful and redundant evidence: some deletions change the strict winner, whereas others leave enough information elsewhere for a selector to recover. The result is deliberately narrower than claiming one unique human reason for every selection, because the edited documents can be incomplete and one operational fact can occur in several sections.

The historical frozen-v0.4 representation matrix motivates, but does not answer, the revised RQ2. It often changes content, structure, candidate generation, and reranking together. For example, with Qwen held fixed, R2 structured representation improves top-1 over R1 flat metadata, and full skill text behaves differently again. These rows show that representation and selector choice interact, but they cannot establish whether the gain comes from additional facts, explicit labels, reduced dilution, or a field-sensitive selector.

The historical public-gold results complicate the story further. On public-gold, R2 is best for Qwen embedding-only retrieval, but R1 plus Qwen reranking has the highest strict top-1. This likely happens because public skills often contain strong provider, platform, or task cues in names and descriptions. In those cases, flat descriptions are not necessarily weak metadata.

7.2 RQ2a Answer: Facts Matter More Consistently Than Headings

RQ2a removes the attribution ambiguities in the historical matrix by holding reviewed propositions and candidate identities fixed. Its strongest conclusion is the operational-content positive control: fielded representations rise by 42.4, 53.1, and 61.8 percentage points over shared-only for Qwen, BM25, and SkillRouter respectively. A selector cannot resolve the near-neighbour decision when candidate-specific operational facts are absent, and all three architectures can exploit those facts when they are restored.

The representation-form result is more qualified. Explicit headings improve BM25 by 2.1 points over matched flat text, which is statistically detectable but below the frozen three-point practical threshold. The same headings reduce Qwen single-vector top-1 by 6.7 points and have no clear effect on SkillRouter cross-encoder selection. This rejects the simple hypothesis that visible field organisation is

universally beneficial. Field labels interact with the scoring architecture: they can provide useful lexical anchors, consume or distort a pooled semantic vector, or be largely neutral when a cross-encoder jointly attends to the query and candidate.

The semantic field-aware result sharpens this interpretation. Frozen uniform-top-two scoring improves Qwen fielded selection by 6.6 points, but its 0.823 top-1 is effectively the same as Qwen flat at 0.824. The field-aware mechanism therefore recovers the penalty of compressing labelled fielded text into one vector; it does not beat the best matched Qwen serialisation. Descriptive field slices further show that the global rule helps success/verification, boundary, workflow, and use-condition cases while harming dependency/resource and output/artifact cases. The appropriate conclusion is not “structure wins”, but that field-wise use is selectively valuable and a single aggregation policy cannot treat all operational fields as interchangeable.

7.3 RQ2b Answer: Full-Library Accuracy And Compression Trade Off

RQ2b tests a different question from RQ2a. It does not hold candidate content fixed, and it does not assign a causal effect to a particular field. Instead, it asks what happens when native metadata, original full bodies, extracted unlabelled evidence, and field-labelled I3C are used to retrieve from the same 2,433-skill library. The strict endpoint contains 381 intended gold labels. It is a demanding full-library retrieval test, but a strict-label miss is not equivalent to downstream task failure or a proof that the returned alternative is useless. V3 is a prospectively frozen internal re-evaluation of historically explored prompts, so its bootstrap intervals characterise fixed-benchmark stability rather than untouched confirmatory generalisation.

The main result is an accuracy–compression trade-off, not a universal I3C benefit. On public-gold B1 retrieval, I2 full bodies outperform I3C for all three first-stage retrievers: 0.644 versus 0.504 for BM25, 0.686 versus 0.538 for Qwen, and 0.806 versus 0.746 for SkillRouter. I3C simultaneously reduces the selector-visible corpus by 72–76% and recorded document/index time by 64–79% relative to I2. Fixed-Top-20 reranking changes candidate order but does not consistently reverse this pattern: I3C remains below I2 in five of six public retriever–reranker combinations. The failure decomposition makes the pipeline boundary visible: I3C has no lower candidate-miss rate than I2 in any comparison, and a reranker cannot promote a gold that failed to enter its fixed Top-20 list.

B4 prevents an overly convenient explanation of this result. Only 4 of 755 retained I3 items associated with the strict-gold prompt records are visible in the corresponding I1 descriptions after whitespace folding. I1’s competitive result is therefore not explained by literal duplication of the retained I3 operational items in short metadata. Coarse task/category cues, benchmark properties, or changes introduced by evidence selection and serialisation are plausible alternatives, but the present post-hoc census does not causally choose among them. The thesis should not claim that flat metadata is generally sufficient, nor that I3C is generally inferior; it reports the observed V3 trade-off and the limits of the explanation.

7.4 Qwen and SkillRouter Represent Different Accuracy–Cost Points

Although both model families operate on text, the tested conditions are not equivalent embedding systems. Qwen `text-embedding-v4` is a bi-encoder: it embeds queries and candidate representations independently, permits document-vector precomputation, and ranks by cosine similarity. SkillRouter RQ2a uses `SkillRouter-Reranker-0.6B`, a cross-encoder that jointly scores each query–candidate pair. It is therefore able to model direct token-level interactions, but the final relevance score cannot be precomputed from the candidate alone.

This distinction explains the observed trade-off. SkillRouter reaches 0.955 top-1 on flat text, the best condition in the matrix, while Qwen flat reaches 0.824. Qwen’s document construction is an amortisable one-time cost and its online scoring is about 2.87 ms per prompt after vectors exist. SkillRouter requires about 11.10 s per novel prompt for three candidate pairs in the measured environment. Warm pair-score cache hits are useful for verification but are not a realistic substitute for novel-query inference. RQ2a did not test a SkillRouter-native field-aware architecture; the field-aware row uses Qwen component embeddings, while historical SkillRouter-first-stage plus M6 rows are a different two-stage pipeline.

RQ2b uses the same model names in a different pipeline role. Its B1 SkillRouter condition is a native bi-encoder, comparable to Qwen as a cached document-retrieval stage; its B2 SkillRouter condition is a cross-encoder reranker over the persisted Top-20 list. Qwen B2 is an independent reranker over the same fixed lists. The results should therefore not be collapsed into a statement that one “SkillRouter score” or one “Qwen score” is universally better: whether a model embeds a full library, ranks a fixed three-sibling set, or reranks Top-20 candidates determines both what it can recover and what cost it incurs.

7.5 Full Skill Text And Extracted I3C Have Different Failure Modes

The RQ2b result rules out two simplistic prescriptions: “always embed the whole document” and “always extract I3C”. Original full bodies retain detail that helps candidate inclusion on the public-gold endpoint, while I3C is a much smaller selector surface. The best choice in this study depends on whether strict top-1 and Recall@20 or document/index burden are the more binding deployment constraint.

The historical frozen-v0.4 and external SkillRouter-Eval-Core rows remain useful as portability context, but they use different prompts, source artifacts, metrics, and extraction versions. They cannot override the source-grounded V3 full-library comparison. In particular, an external BM25 result where an earlier I3C extraction is favourable does not establish that I3C must be favourable for every library or first-stage retriever.

7.6 Lexical Baselines Are Stronger Than Expected

The historical M6-v1 local runs show that lexical candidate generation remains a serious baseline. Controlled TF-IDF/schema plus M6-v1 field-aware reranking reaches 69.8% top-1 and 97.1% top-5. Public-gold TF-IDF/flat plus task-field reranking reaches 68.1% strict top-1.

This result changes the framing of the thesis. The project should not be presented as a simple contest between dense embeddings and structured representations. A more accurate framing is:

Which operational selection information should be exposed, and which retrieval or reranking strategy can use it reliably under semantic confusability?

In the current benchmark, lexical retrieval can exploit strong names, descriptions, output words, and workflow terms. That does not make the benchmark invalid, because real systems also contain lexical cues. But it means the final analysis must treat lexical+field-aware methods as serious baselines and investigate whether their wins come from genuine procedural matching or from surface cue overlap.

7.7 What Information Appears Useful

The controlled field-isolation suites suggest that output artifacts, input conditions, preconditions, workflow steps, procedural cues, use conditions, boundary evidence, dependency/resource constraints, and

concrete success criteria can all function as retrieval evidence. These fields encode procedural suitability: what the skill is for, what it expects, what it produces, how it operates, what scope or authority limits apply, what external capability it requires, and how completion is judged.

The use-condition result is the most direct version of this claim because the field states when the skill should be selected. It is clearly discriminative in the controlled clusters, but the paraphrase rows show that it is not merely a perfect direct hint. BM25 drops more sharply under paraphrase than Qwen embedding, which suggests that task/use-condition matching sometimes requires semantic interpretation of the user's intent rather than only lexical overlap with the skill's trigger wording.

The boundary/not-for result also shows why not all fields should be treated as the same kind of evidence. Explicit boundary prompts produce strong gains, but implicit-authority prompts are much harder. When the user asks for an internal review note, triage packet, or evidence summary, a human may infer the authority limit, but ordinary lexical and embedding retrievers often do not. Boundary information is therefore better understood as a conditional scope or guardrail signal than as a simple positive matching field.

The dependency/resource result gives a second example of conditional evidence. In controlled clusters, capability compatibility is highly discriminative: when the skill-side field states a required platform, tool, API, runtime, permission, version, or resource, both BM25 and Qwen embedding can use it to break near-neighbour ties. The difficult part is acquisition and normalisation. A user may not say "React 18", "Chrome DevTools MCP", or "GitHub token" in the natural request. The router may need to obtain that cue from repository files, configured tools, credentials, package metadata, or runtime state. Public skills also mix hard requirements with generic setup text: `npm install`, `npm ci`, or ordinary build/test commands are usually execution preparation rather than identity evidence, while Chrome DevTools MCP, `GH_TOKEN`, or Node.js 20+ are capability constraints. Therefore the thesis should not claim that raw dependency prose is always useful. It should claim that specific external capability compatibility is useful when the selector has access to sufficient capability context, whether that context comes from the request or from the routing environment.

Success/verification is a third conditional field type. It is useful when the request names the acceptance gate directly, but it is much less robust under paraphrase: in the completed suite, both BM25 and Qwen embedding fall to 48.0% top-1 on paraphrased success criteria. This does not make the field useless. It shows that success criteria are often semantically close to one another, such as schema validation,

evidence anchoring, threshold checks, and smoke checks. The field is therefore better interpreted as a completion-gate or late-stage tie-breaker signal than as a strong first-pass discriminator.

The public-gold and public-audit evidence is more cautious. Public descriptions can already contain strong provider, platform, task, or artifact cues, and public skill bodies often mix useful routing evidence with setup text, examples, broad parent instructions, and duplicated capabilities. This does not mean operational fields are useless. It means the controlled suites provide the clearest evidence that exposing a field can help, while the public experiment tests whether removing the same kinds of information matters in less controlled documents.

The most defensible current claim is therefore not that every field always improves retrieval. The current claim should be:

Candidate-specific operational information can improve routing when it is visible and matches the request, but the size and reliability of that improvement depend on the field, prompt, retriever, and surrounding document.

RQ1 is not a competing retrieval architecture. SSL proposes a complete layered skill representation, while SkillRouter and SkillRet evaluate larger retrieval pipelines [2, 10, 23]. RQ1 asks a narrower question: when several skills are plausible, which kinds of operational information can change the selection? The controlled addition and public removal experiments test that question separately from architecture. They therefore complement these systems rather than claiming to outperform them; representation and retrieval architecture are compared under RQ2.

7.8 Why Existing Field-Aware Prototypes Are Diagnostic

M6-v1-local is useful because it operationalises the thesis claim: it parses request-side cues and compares them with skill-side fields. It also separates first-stage candidate recall from reranking quality. This helps diagnose whether failures come from the retriever excluding the gold skill or from the reranker ordering candidates poorly.

But M6-v1-local is not yet a final robust structure-aware method. It is still largely lexical. It can miss semantically equivalent wording, overvalue repeated field terms, and under-handle contradiction, negation, hierarchy, and broad parent skills. Its public-gold weakness is therefore informative. It shows that

field-aware retrieval needs semantic matching or LLM-assisted field interpretation, not only manually designed field names.

M6-v2 already provides a semantic field-matching prototype, but it remains tied to the old frozen-v0.4 representations, first-stage blending, selected field sets, request parsing, penalties, and calibration. The completed RQ2a adapter reuses M6-v2's embedding and caching components while removing first-stage blending, adding all seven fields, hiding target-field metadata, and freezing one target-agnostic aggregation rule on development clusters. Graph, tree, and DAG variants are outside the completed RQ2a scope.

7.9 Benchmark Validity

The benchmark has several strengths. It contains controlled confusable clusters, a large 2433-skill scale condition, generated background skills, imported public skills, public-gold prompts, acceptable alternatives, and validation checks for integrity, procedural distinctness, semantic confusability, and leakage.

The benchmark also has limitations. The controlled skills were created with AI assistance under a researcher-defined rubric and then checked and approved by the researcher. They may therefore express procedural distinctions more cleanly than publicly available skills, and some gold field values are longer or more detailed than their alternatives. The public experiment uses purposefully selected, removal-eligible groups rather than a random sample. It tests less controlled documents, but its result is still limited by group selection, field-specific eligibility, broad parent skills, duplicate capabilities, provider-specific names, and confirmation of the frozen gold labels by the same researcher with the labels visible.

The external SkillRouter-Eval-Core benchmark has a different limitation. It is much larger and externally sourced, but its scored tasks are more informative than the local controlled prompts. The median scored task is 169 rough tokens, and most tasks include file/path references plus explicit input and output cues. Its skill bodies are also highly markdown-normalised, with frequent headings such as `overview`, `when to use`, `workflow`, `quick start`, and `dependencies`. This can make full-body retrieval more attractive than it would be in a shorter, near-neighbour controlled setting. The external benchmark therefore strengthens portability evidence, but it does not remove the need for the local controlled benchmark.

The benchmark should therefore be described as a stress test for semantic confusability, not as a universal public benchmark. Its value is that it isolates a specific failure mode: skills that look semantically plausible but differ procedurally. Its limitation is that final generalisation requires more public-skill validation and downstream testing.

7.10 Failure Modes

The current experiments reveal several recurring failure modes:

- **First-stage exclusion:** the gold skill is not included in the candidate set, so reranking cannot recover it.
- **Object-procedure confusion:** the selector focuses on the shared task object, such as PDF, GitHub, or spreadsheet, rather than the required procedure.
- **Full-document dilution:** full skill text contains the right information but also enough unrelated text to weaken embedding similarity.
- **Provider cue dominance:** public skill names or providers become strong cues, sometimes correctly and sometimes superficially.
- **Boundary failure:** negative or avoid conditions are treated as ordinary positive text or ignored.
- **Broad parent selection:** a broad public skill is selected instead of a more specific atomic skill.
- **Lexical overfitting:** a method ranks the correct skill because of surface term overlap rather than deeper procedural compatibility.
- **Underspecification:** the user request lacks enough evidence to choose a stable skill without clarification.

7.11 Implications for Skill Retrieval Systems

The practical implication is that scalable skill libraries should not rely only on names and short descriptions, but they also should not assume that full-document embedding solves the problem. A better architecture should separate first-stage candidate generation from final selection, preserve procedural selection information, and use field-aware logic for outputs, workflows, preconditions, dependencies, and boundaries.

The current results also suggest that multiple retrieval strategies may be economically useful. Lexical retrieval is cheap and strong when skill names and descriptions are well written. Dense retrieval is useful for broader semantic recall but can be sensitive to representation length and noise. Reranking improves ordering, but its value depends on whether the candidate set contains the gold skill and whether the reranker sees the right information.

Cost should therefore be analysed as an amortised pipeline rather than as a single per-query number. RQ2b measures the selector-visible corpus, B1 document/index work, B1 query work, and fixed-Top-20 reranking work. It does not contain a durable comparable I3C extraction wall-time or provider invoice, so it cannot calculate a true break-even query count. The appropriate claim is narrower: I3C lowers recorded B1 representation/index burden relative to I2, while the missing construction-cost term and the observed strict-label loss must both remain visible in any deployment decision.

7.12 Current Limitations

RQ2a is complete, but its scope is deliberately narrow. It is a fixed-candidate mechanism study in which the gold skill is always present among three controlled siblings. RQ2b complements it with full-library candidate generation, Recall@20, fixed-Top-20 reranking, and a public-gold stratum, but it cannot isolate the effect of one field because the representations differ in content and extraction. In the controlled suites, the candidate text is identical before the target field is added, so the observed change measures the effect of showing that field content in that suite. Because the field values were not matched for length or detail, the experiment does not separate field meaning from wording or amount of information. The V3 public and controlled full-library sources provide strict-label retrieval evidence under a larger, messier candidate pool.

The selector coverage also bounds the claim. The Qwen field-aware condition uses one frozen uniform-top-two rule; field-level slices are descriptive and cannot justify post-hoc field-specific weights. No SkillRouter-native field-aware condition was tested. The RQ2b availability census is post-hoc and provenance-bounded: it shows where retained evidence occurs, not which information caused a retrieval result. Its strict labels are intended routing targets rather than an acceptable-alternative or downstream-success endpoint. Automatic source/provenance validation does not establish independently reviewed semantic completeness of I3C: a protocol-invalid manual review was quarantined and no replacement

semantic-completeness audit was used. The cost measurements are implementation- and hardware-specific, and RQ2b lacks a complete I3C construction ledger. M4 tree routing, M5 graph retrieval, full-scale progressive disclosure, and downstream task success remain future or separate studies rather than missing rows from the completed RQ2a/RQ2b evidence.

Conclusion

This thesis investigates operational information and representation strategies for scalable agent skill retrieval. The motivating problem is semantic confusion: as skill libraries grow, many skills can sound relevant while requiring different inputs, outputs, workflows, dependencies, or boundaries. In these cases, the correct skill may exist in the library, but the selector may surface a plausible yet procedurally inappropriate alternative.

The thesis uses a layered vocabulary for skill artifacts, skill representations, and skill retrieval. This vocabulary is useful because a full skill artifact may contain rich procedural information, while the selector may see only a compressed card. Conversely, simply exposing more text does not guarantee better retrieval if the relevant signal is diluted.

The earlier frozen-v0.4 benchmark contains 2433 skills, 245 controlled prompts, 144 public-gold prompts, and 12 low-information stress prompts. Its representation, SkillRouter, and M6 matrices remain exploratory evidence that representation and selector choice interact. The thesis-facing full-library result instead uses the later source-grounded V3 strict-gold endpoint: 2,433 skills and 381 prompts, comprising 243 controlled and 138 public-gold prompts.

The exploratory evidence also shows that lexical baselines are strong. This prevents the thesis from being framed as a simple dense-embedding-versus-structure comparison; the full-library result similarly requires each representation to be compared under the same first-stage retriever and candidate set.

The controlled answer to RQ1 is that all seven tested operational information types can help distinguish near-neighbour skills when the candidate-specific field is visible. The result is not a ranking: each field uses a different 50-cluster suite, and robustness changes with prompt wording and retriever. Input requirements remain robust under paraphrase, Qwen handles paraphrased output requirements better than BM25, and implicit boundaries, paraphrased success criteria, and fine-grained workflows are harder for

both. Dependency information is useful when it expresses concrete capability compatibility rather than generic setup commands. Examples/tests provide a negative control: generic examples do not reliably distinguish capabilities, although near-exact request/example matches can strongly attract a selector.

The field set was not selected from the controlled benchmark alone. A separate model-assisted binary review of 29,292 frozen public primary skill documents found each candidate type in 64.91–83.34% of complete documents, with use condition in 80.68%, output/artifact in 79.76%, workflow/procedure in 83.34%, and dependency/resource in 75.32%. This full-document census includes native names and descriptions as well as Markdown content, and therefore supports only the descriptive claim that these information types are common in the sampled public skill ecosystem. Together with the literature review, it provides the basis for proposing the RQ1 fields; the controlled and source-removal experiments, rather than this prevalence count, establish their routing relevance.

The public original-document experiment qualifies this answer. In the purposefully selected, removal-eligible public groups, removing cited use-condition and success/verification information lowers strict Hit@1 under both BM25 and Qwen, while input, output, workflow, and dependency effects are stable only for BM25; boundary/not-for has no stable positive Top-1 effect. Group removal of task-specification or execution/verification information lowers Hit@1 for both retrievers. These groups therefore contain both useful and redundant evidence. The edit may leave a skill incomplete, so this result concerns routing after information removal rather than execution. All controlled clusters, public gold labels, and scored removals were confirmed by the researcher with the labels or edits visible; this was not independent or blinded annotation.

RQ2a now has a confirmatory answer. Across 280 untouched clusters, 600 prompts, 22 conditions, and 13,200 aligned fixed-candidate decisions, operational facts are essential but field headings are not universally beneficial. Relative to matched flat propositions, fielded text improves BM25 by 2.1 percentage points, reduces Qwen single-vector top-1 by 6.7 points, and changes SkillRouter cross-encoder top-1 by only -0.4 points. Frozen Qwen field-aware scoring recovers 6.6 points over fielded single-vector selection, but its 0.823 top-1 is essentially equal to Qwen flat at 0.824. SkillRouter is the strongest selector, reaching 0.955 on flat text, but its joint query–candidate cross-encoder inference is substantially more expensive than cached bi-encoder or lexical scoring.

The bounded answer to RQ2a is therefore that representation value depends on retrieval architecture and field type. Preserving operational facts matters consistently; merely displaying headings does not.

Explicit field-wise scoring is valuable for some conditional fields and can repair pooled-vector loss, but the frozen uniform-top-two rule is not a universal replacement for a strong flat representation.

RQ2b compares I1 native metadata, I2 original full bodies, I3-flat evidence, and field-labelled I3C under three first-stage retrievers and two fixed-Top-20 rerankers. It finds an accuracy–compression trade-off: on public-gold B1, I2 exceeds I3C under BM25 (0.644 versus 0.504), Qwen (0.686 versus 0.538), and SkillRouter (0.806 versus 0.746), while I3C reduces selector-visible corpus tokens by 72–76% and measured document/index time by 64–79%. I3C also remains below I2 in five of six public fixed-list reranking configurations. No representation universally dominates.

Only 4 of 755 retained I3 items associated with strict-gold prompt records are description-visible. Literal duplication therefore does not explain I1’s competitive performance, while other causal explanations remain untested. The strict-label endpoint and provenance/availability checks do not establish unique task correctness, field-level causality, independently reviewed semantic completeness of I3C, or complete construction cost.

The contribution is not a universal skill-router benchmark or a provider leaderboard. It is a field-isolated benchmark and information-centred empirical account of why semantically similar skills are difficult to retrieve, which operational information appears to matter, and how representation and retrieval strategies can preserve, organise, exploit, or fail to use that information. RQ2a establishes the controlled mechanism result; RQ2b establishes the full-library accuracy–compression trade-off under source-grounded strict-label evaluation. Representation and retrieval must therefore be evaluated together rather than ranked as independent ingredients. Tree/graph retrieval and downstream execution remain future work.

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Benchmark Rubric

This appendix summarises the benchmark validation loop used in the project. The full operational rubric is maintained in `skill_benchmark/notes/benchmark_methodology_rubric.md`.

A.1 Validation Steps

- (1) **Skill and prompt integrity.** Every prompt has a unique identifier, every gold skill exists, every listed alternative exists, and skill metadata can be parsed.
- (2) **Procedural requirement alignment.** Each prompt must contain concrete requirements that justify why the gold skill is better than its alternatives.
- (3) **Semantic confusability.** Alternatives should be plausible neighbours, not random unrelated distractors.
- (4) **Gold-label stability and leakage.** Prompts should not reveal the answer through exact skill names, copied phrases, or hidden assumptions.
- (5) **Scale regime.** The benchmark should include enough skills to create retrieval pressure and context pressure.
- (6) **Public-skill field audit.** Public skills should be checked for observed, extractable, and proposed representation fields.
- (7) **Public-gold validation.** Public skills can be used as gold targets only after atomisation, leakage checks, semantic alternatives, and manual adjudication.
- (8) **Offline selector evaluation.** Methods should be compared on top-1, top-k, MRR, non-core false positives, and candidate budget.
- (9) **Progressive-disclosure baseline.** The normal main-agent skill-selection pattern should be evaluated where feasible.
- (10) **Downstream validation.** A smaller sample should test whether retrieval differences affect final task artifacts.

A.2 Historical Frozen-v0.4 Pass/Warning Summary

This table records the historical frozen-v0.4 validation state. Its 245/144 prompt counts and acceptable-alternative status are not the thesis-facing V3 RQ2b strict-only endpoint.

Step	Current status
Integrity	PASS: 245 controlled prompts, 144 public-gold prompts, and 2433 skills resolve.
Procedural distinctness	PASS: controlled 245/245 prompts and 830/830 pairs; public-gold 144/144 prompts and 575/575 pairs.
Prompt-specific alignment	PASS with caveats: controlled 238/245 prompts; public-gold 131/144 prompts.
Semantic confusability	PASS: controlled 243/245 prompts; public-gold 144/144 prompts.
Prompt leakage	PASS: 0 critical and 0 high-risk leaks in both main strata.
Scale	PASS: 2433 skills; R1 approximately 333k visible tokens.
Public field audit	Historical frozen-v0.4 status: 460/460 imported public skills audited. The current prevalence evidence is a separate 29,292-document full-primary-document census.
Public-gold	Historical frozen-v0.4 PASS with separate strict and acceptable-alternative scoring; not used for V3 strict-only results.
Downstream validation	Planned but not executed.

TABLE A.1: Historical frozen-v0.4 validation summary.

APPENDIX B

Prompt and Skill Examples

This appendix gives representative examples of the intended benchmark pressure. The exact prompt files are stored under `skill_benchmark/prompts/` and `skill_benchmark/prompts_public_gold/`.

B.1 PDF Table Extraction

User request: extract a table from a PDF while preserving row and column structure.

Gold skill: PDF layout table extractor.

Plausible alternatives: PDF question answerer, PDF OCR cleaner, document field extractor, PDF-to-DOCX converter.

Procedural distinction: the gold skill must inspect layout and reconstruct table structure. A PDF question-answering skill may ground answers in pages but not reconstruct tables. OCR cleaning handles scanned text uncertainty. Conversion changes format but is not optimised for table reconstruction.

B.2 GitHub CI Root Cause

User request: inspect a failed CI run and identify the likely root cause from logs.

Gold skill: CI log root-cause debugger.

Plausible alternatives: deployment build triager, PR review-comment resolver, repository code reviewer, release changelog generator.

Procedural distinction: the gold skill must parse CI logs and isolate the failing command, error, dependency, or environment condition. Review-comment resolution and code review are semantically adjacent GitHub workflows but require different inputs and outputs.

B.3 Observability Trace Diagnosis

User request: diagnose latency across a distributed trace path.

Gold skill: distributed trace investigator or implicit trace path diagnoser, depending on the prompt stratum.

Plausible alternatives: latency anomaly detector, SLO breach checker, service dependency mapper, metrics overview.

Procedural distinction: trace diagnosis follows spans and service boundaries. A metrics overview summarises signals, and a dependency mapper describes architecture, but neither necessarily reconstructs a trace path.

B.4 Public-Gold Example: Figma Implementation

User request: implement an existing Figma node as production UI code with close visual parity.

Gold skill: public Figma implement-design skill.

Plausible alternatives: broad Figma parent skill, Figma generate-design skill, generic frontend implementation skill.

Procedural distinction: the gold skill targets an existing Figma node and produces code. A broad parent skill may route to several subskills, while generate-design writes or creates design content rather than implementing an existing node.

B.5 Public-Gold Ambiguity Example

Some public-gold prompts are not stable strict targets. For example, a public OCR skill may be valid for scanned PDF OCR, but a controlled OCR-cleaning skill may be better if the prompt asks for uncertain recognition regions and page anchors. These cases are either revised, excluded, or marked with acceptable alternatives before final reporting.

B.6 Public-Document Field-Presence Review

The descriptive review of 29,292 public primary skill documents was completed in 31 batches by a Codex `gpt-5.6-terra` agent using `xhigh` reasoning. The operational instruction was:

For each candidate information type, record 1 only when that information is expressed somewhere in the complete primary skill document, and otherwise record 0. Read the YAML name and description together with every Markdown section, table, and example. Do not inspect attached scripts, references, or other auxiliary files. Judge presence only; do not judge document quality, routing usefulness, or whether the field is required by every skill.

This is a reconstruction of the recorded decision rule, not a verbatim preserved batch prompt. The retained evidence contains the aggregate summary and 31 batch totals, but not the 29,292 row-level model decisions. The prevalence result is therefore used only to show that the candidate information types recur in the sampled public documents.

B.7 RQ1 Intervention Examples

These examples show the exact type of change made by each RQ1 experiment. They are included for interpretability and auditability, not as additional evidence. To avoid choosing unusually favourable outcomes, the controlled example is the first row in the approved controlled-review ledger. The public example is the approved use-condition transformation with the fewest blanked non-empty lines; it was selected before inspecting its retrieval scores.

B.7.1 Controlled field addition

Controlled unit `uc01_api_contract_review` contains three API-work skills. Its paraphrased request is:

Check the already-written endpoint specification for compatibility and contract-quality concerns.

When the target field is hidden, all three candidates receive exactly the same serialised shared context. This includes the same broad description, input, output, workflow, success, boundary, dependency, and generic use-condition text. The candidate documents are therefore tied by construction. When the field is shown, one candidate-specific line is appended to each otherwise identical representation:

Gold skill.: Use Condition: Use when the user needs review of an existing API contract or endpoint specification.

Alternative 1.: Use Condition: Use when the user needs planning for a new external API integration.

Alternative 2.: Use Condition: Use when the user needs design of a new webhook receiver workflow.

Across the direct and paraphrased requests for this frozen cluster, tie-aware Top-1 is .333 for both BM25 and Qwen while the field is hidden, and 1.000 for both after the field is shown. The complete result uses all 50 use-condition clusters rather than this single example.

B.7.2 Public information removal

Public composition `OR82-055` contains three naturally authored onboarding skills: a customer-account onboarding plan, an in-product user-activation flow, and an employee 30/60/90-day onboarding plan. For public-gold case `P-0129`, the in-product activation skill is the reviewed gold. Its direct request asks for a redesign of the self-serve journey between signup and the action that predicts retention, including drop-off analysis, in-product guidance, follow-up messages, measurement, and experiments.

In the intact gold document, the frontmatter description begins:

When the user needs to design, improve, or audit a post-signup activation flow to get new users to their first value moment.

Under `REMOVE_USE_CONDITION_R3`, that description and the cited lines in the document's `When to Use` section are replaced with blank lines. No remaining text is rewritten. The same rule is applied to all three candidates: four, three, and five non-empty lines respectively are blanked, while document identity, line count, prompt, candidate order, and gold label stay fixed.

For the direct request, BM25 still selects the gold after removal, but its native gold-versus-best-alternative score gap falls from 17.444 to 14.313. Qwen also keeps the gold at rank one, while its cosine gap changes from .158 to .167. The two score scales are not comparable. This single case deliberately illustrates that removal can weaken one retriever without changing Top-1 and need not move every retriever in the same direction; the RQ1 conclusion comes from the paired aggregate over all eligible groups.